sss

Program Goals and Prerequisites

- Goals
  - Provide you with an introduction to the techniques and methodologies used to produce digital Very Large Scale Integration designs
  - Give you some hands-on experience with the SPICE for circuit simulation and EDA tools for design and hardware description language and programmable logic devices for Computer Aided Design (CAD)
  - Create a foundation for further exploration of digital VLSI design
- Prerequisites
  - Basic understanding of circuits and electronics
  - Basic understanding of logic design



What do you need to know to design a modern digital integrated circuit?

- Semiconductor processing
- Transistor physics
- Circuit design
- Logic design
- Signaling and systems engineering
- ECAD
- Programming
- Plus application specific knowledge



Module 1

Overview of VLSI:
Complexity, Design Methodology

Dr. Manisha Pattanaik



Lecture Outline

- Historical development of computers
- Introduction to a basic digital computer
- Five classic components of a computer
- Microprocessor
- IC design abstraction level
- Processor Trend
- Developmental trends of ICs
- Moore's Law



Introduction to Digital Circuits



What is a digital Computer ?

A fast electronic machine that accepts digitized input information, processes it according to a list of internally stored instruction, and produces the resulting output information.

List of instructions → Computer program

Internal storage → Memory

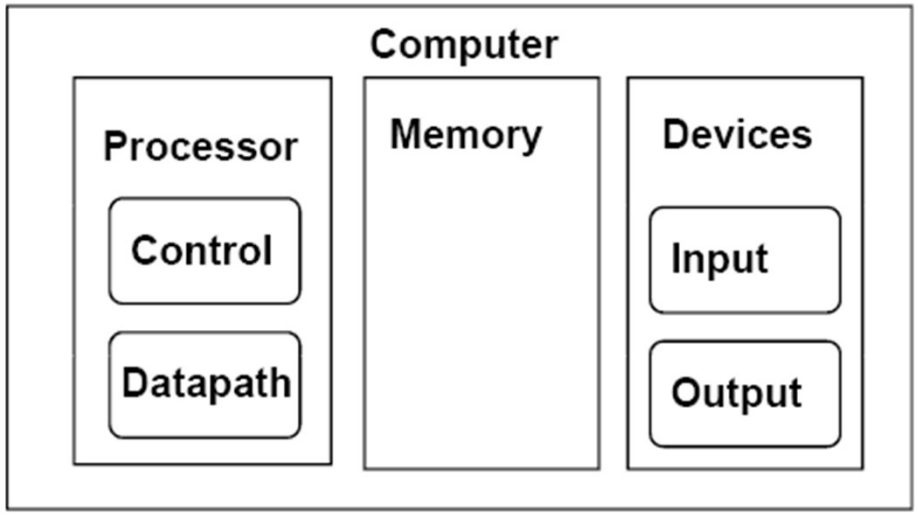


Different Types and Forms of Computer

- Personal Computers (Desktop PCs)
- Notebook computers (Laptop computers)
- Handheld PCs
- Pocket PCs
- Workstations (SGI, HP, IBM, SUN)
- ATM (Embedded systems)
- Supercomputers



Five classic components of a Computer



((1) Input, (2) Output, 3) Data path, 4) Controller, and (5) Memory

What is a microprocessor ?

- A microprocessor is an integrated circuit (IC) built on a tiny piece of silicon. It contains thousands, or even millions, of transistors, which are interconnected via superfine traces of aluminum. The transistors work together to store and manipulate data so that the microprocessor can perform a wide variety of useful functions. The particular functions a microprocessor performs are dictated by software. (source : Intel)
- Simply speaking, microprocessor is the CPU on a single chip. CPU stands for "central processing unit" also known as processor.
- Processor can be "general purpose" or "special purpose". A special purpose processor is also known as "application specific integrated circuit" (ASIC).

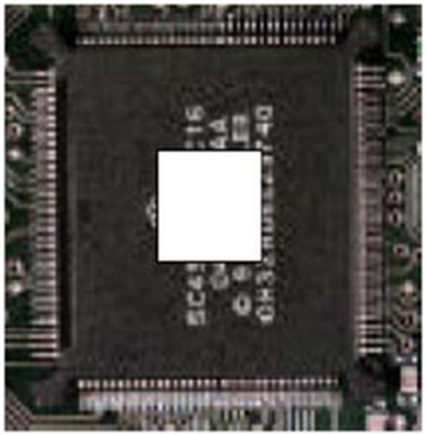


What is an Integrated Circuit ?

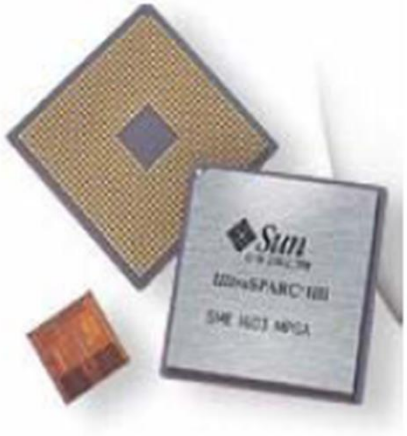
- An integrated circuits is a silicon semiconductor crystal containing the electronic components for digital gates.
- Integrated Circuit is abbreviated as IC.
- The digital gates are interconnected to implement a Boolean function in a IC .
- The crystal is mounted in a ceramic/plastic material and external connections called “pins” are made available.
- ICs are informally called chips.



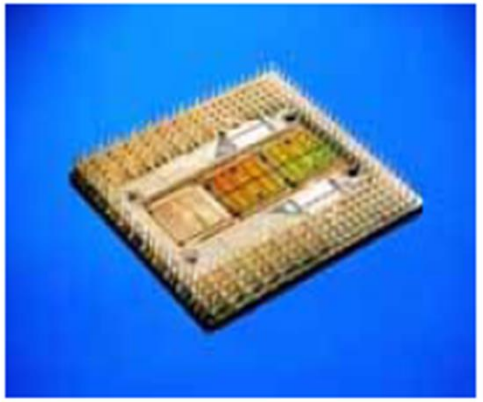
How does a microprocessor look?



(1) ASIC



(2) Sun UltraSparc



(3) PentiumPro



Historical Development



VLSI Technology: Highest Growth in History

- 1958: First integrated circuit
  - Flip-flop Flip using two transistors
  - Built by Jack Kilby at Texas Instruments
- 2003
  - Intel Pentium 4 micro processor (55 million transistors)
  - 512 Mbit DRAM (> 0.5 billion transistors)
- 53% compound annual growth rate over 45 years
  - No other technology has grown so fast so long
- Driven by miniaturization of transistors
  - Smaller is cheaper, faster, lower in power!
  - Revolutionary effects on society



Invention of the Transistor

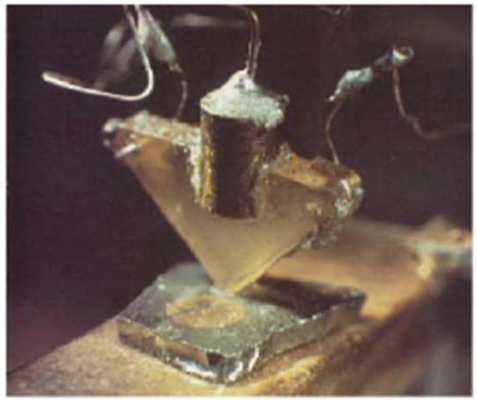
- Invention of transistor is the driving factor of growth of the VLSI technology
- Vacuum tubes ruled in first half of 20th century

Large, expensive, power-hungry, unreliable

- 1947: first point contact transistor

– John Bardeen and Walter Brattain at Bell Labs

– Earned Nobel prize in 1956





Transistor Types

Bipolar transistors

– n p n or p n p silicon structure

– Small current into very thin base layer controls large currents between emitter and collector

– Base currents limit integration density

• Metal Oxide Semiconductor Field Effect

Transistors (MOSFET)

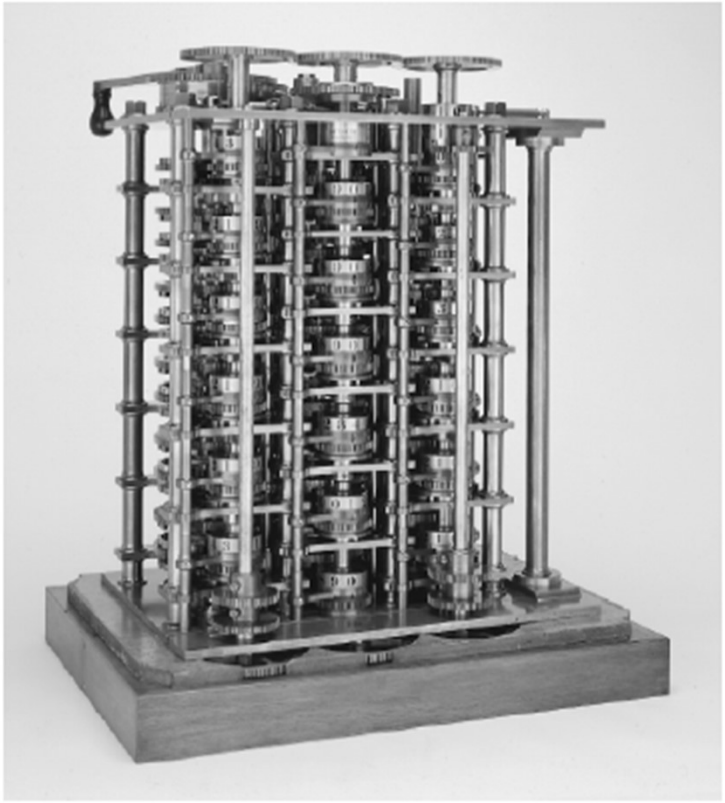
– nMOS and pMOS MOSFETS

– Voltage applied to insulated gate controls current between source and drain

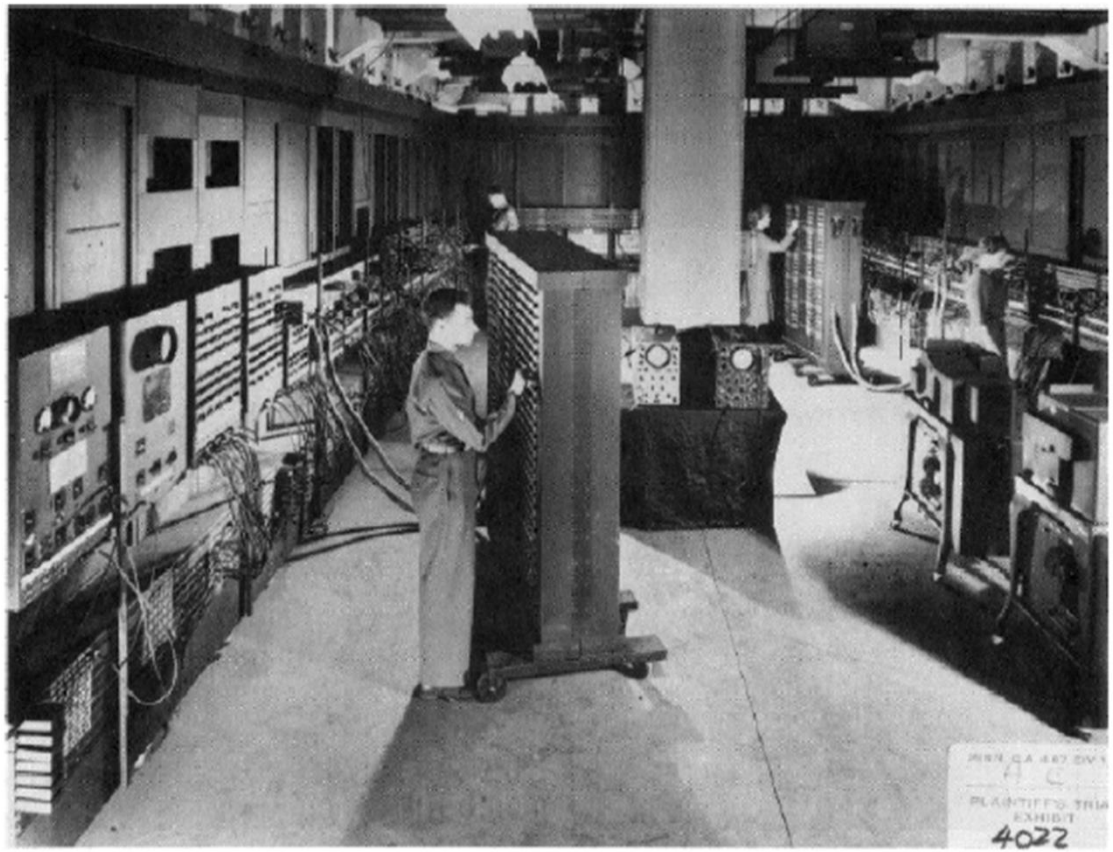
– Low power allows very high integration



The Babbage Difference Machine in 1832



The First Electronic Computer in 1946 (ENIAC)



From

- 1945

  ENIAC filled an entire room!

  17,468 vacuum tubes,

  70,000 resistors, and

  10,000 capacitors

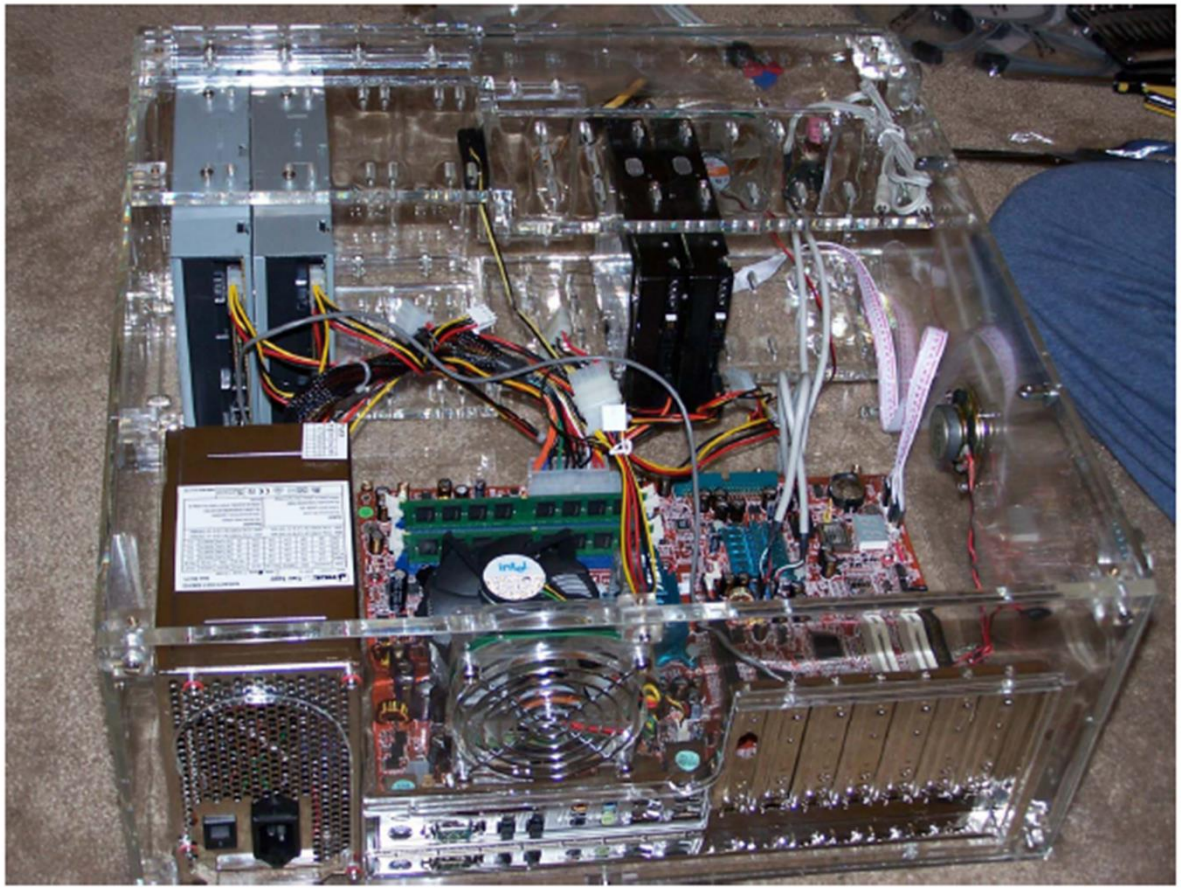
  6,000 manual switches

  and many blinking lights!

  Could add 5,000 numbers in a single second

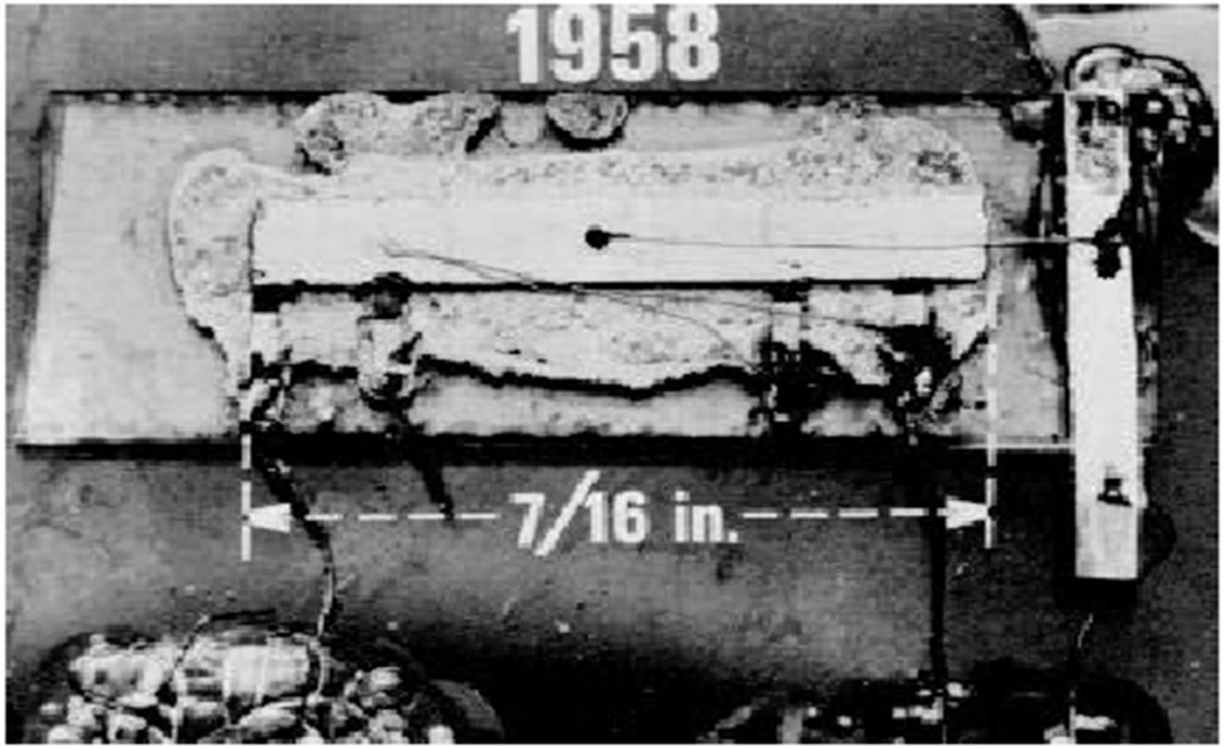


How a Home PC Looks Today??



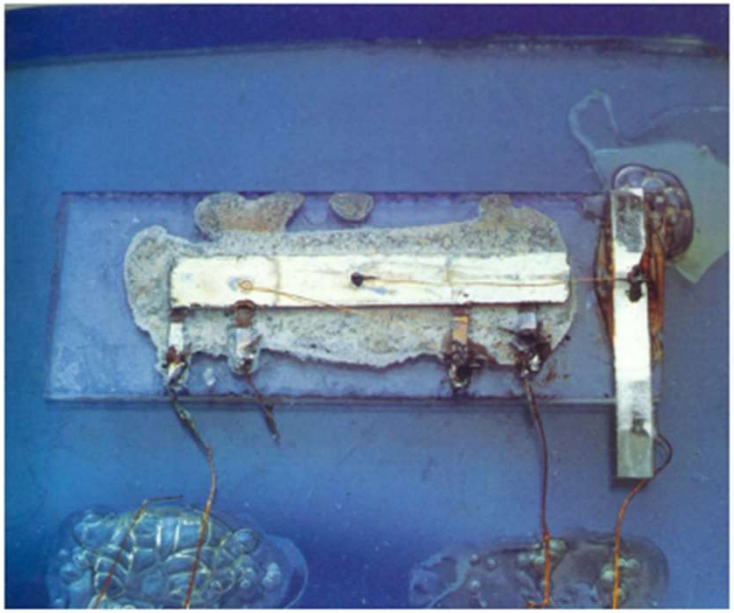
1958

A device having multiple electrical components and their interconnects manufactured on a single substrate



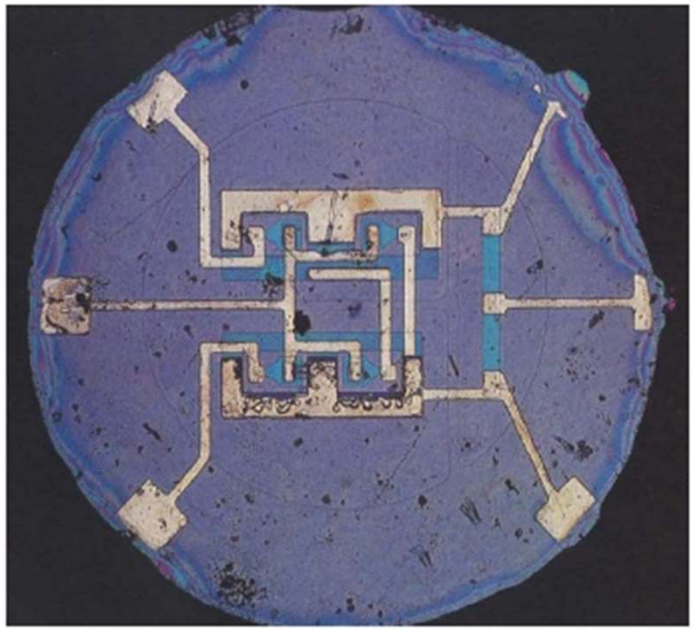
First Integrated Circuit - 1958

- The First Integrated Circuit – Jack Kilby, Texas Instruments
- 1 Transistor and 4 Other Devices on 1 Chip
- Winner of the 2000 Nobel Prize



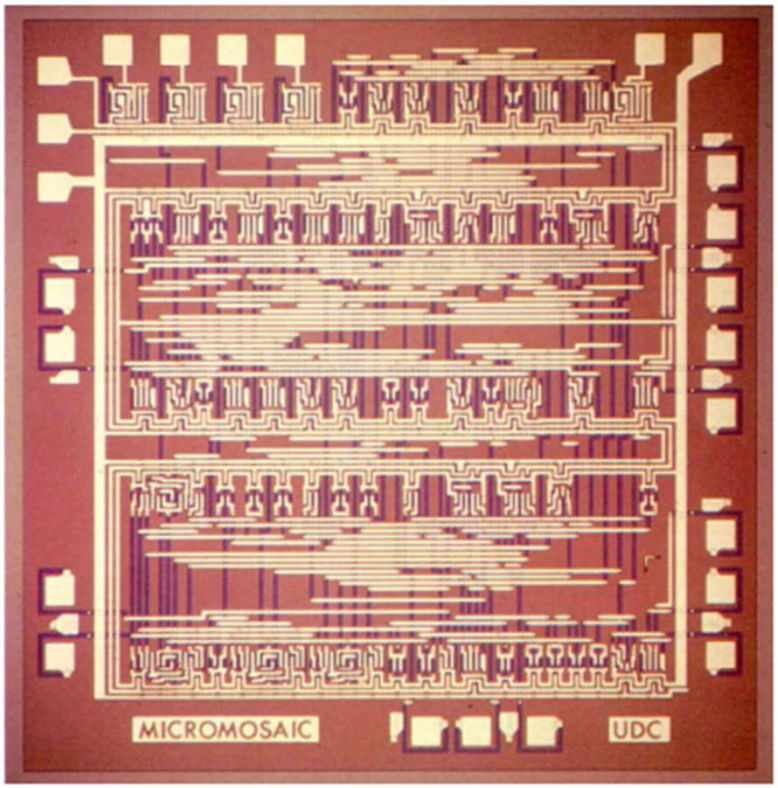
First Commercial Planar IC - 1960

- Fairchild -- One Binary Digital (Bit) Memory Device on a Chip
- 4 Transistors and 5 Resistors
- Start of Small Scale Integration (SSI)!! We are in VLSI!!





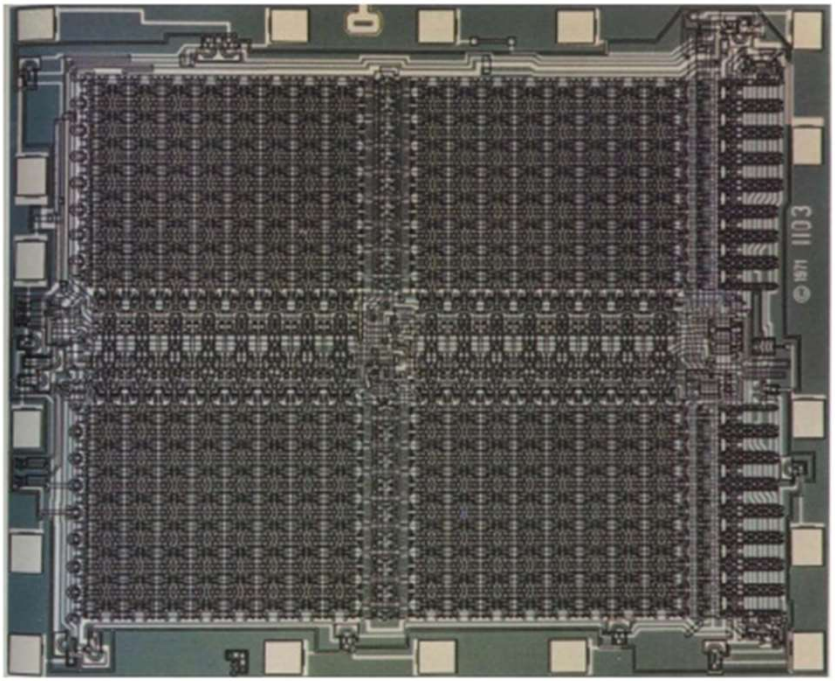
First IC Created with Computer-Aided Design Tools -- 1967



μMOSAIC – Fairchild



First 1,024 Bit Memory Chip -- 1970



Intel Corporation DRAM

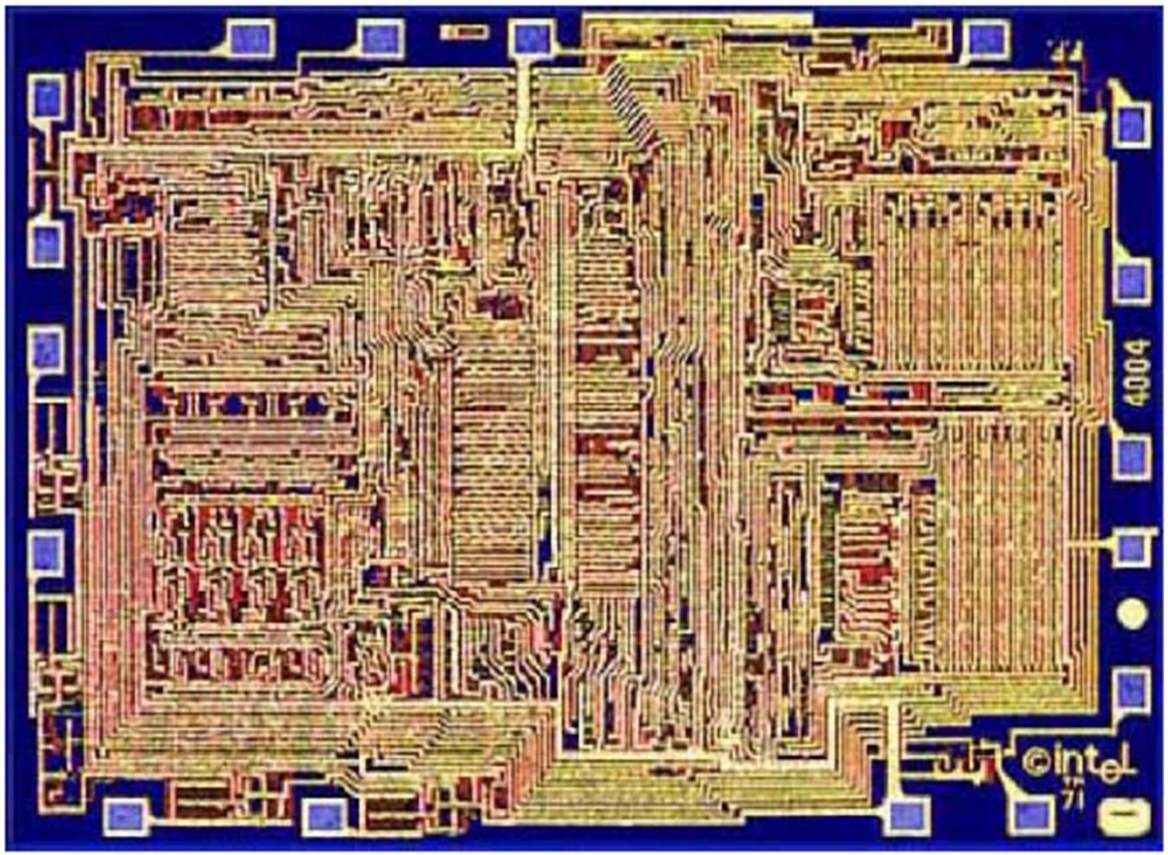
- 1970's processes usually had only nMOS transistors

– Inexpensive, but consume power while idle.

- 1980s-present: CMOS processes for low idle power

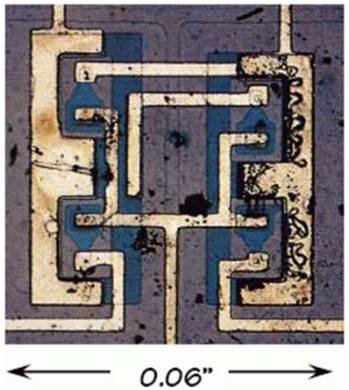


Intel 4004 : 2.3K Transistors (1971)



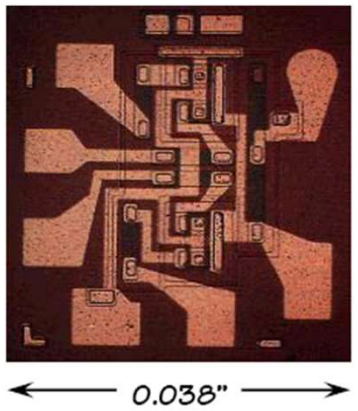


1960's: era of integration (social and electrical!)



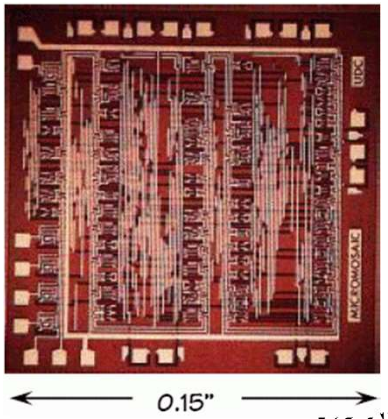
1961: TI and Fairchild introduced the first logic IC's (cost ~$50 in quantity!). This is a dual flip-flop with 4 transistors.

1963: Densities and yields are improving. This circuit has four flip flops.



1966: Robert Dennard invents 1-T DRAM at IBM TJ Watson Research Center.

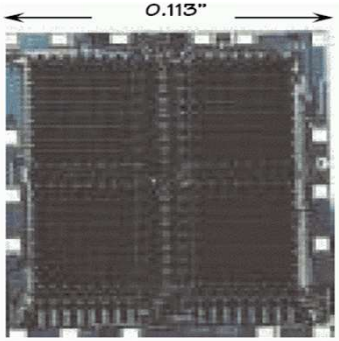
1967: Fairchild markets this semi-custom chip. Transistors (organized in columns) could be easily rewired using a two-layer interconnect to create different circuits. This circuit contains ~150 logic gates. Masks are laid-out, cut and checked by hand… beginnings of a design flow but no computer automation.



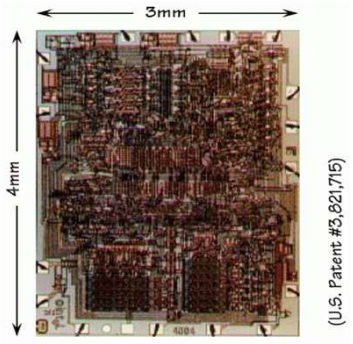


INTegrated ELectronics = Intel

1968: Noyce and Moore leave Fairchild and found Intel. No business plan, just a promise to specialize in memory chips. They and Art Rock raise $2.5M in two days and move to Santa Clara. By 1971 Intel had 500 employees; by 1983 it had 21,500 employees and $1.1B in sales.



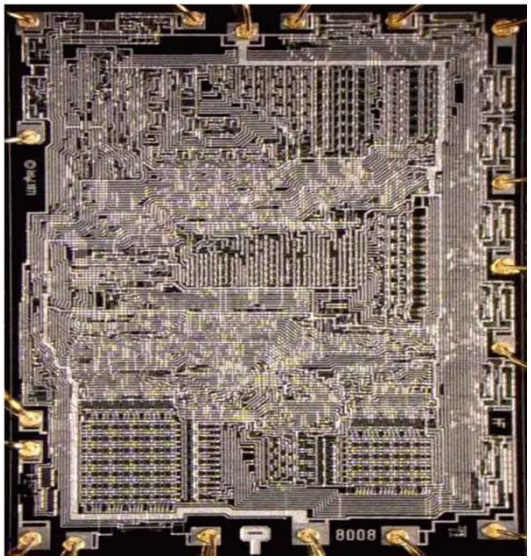
In 1970, making good on its promise to its investors Intel (Joel Karp, Les Vadasz, John Reed) starts selling a 1K bit PMOS RAM, the 1103. It was a bear to interface to, but its density and cost make it the only game it town. Core memory dies…



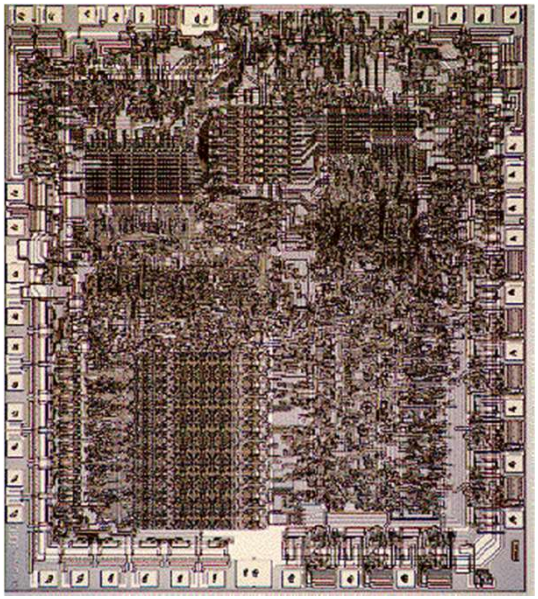
In 1971 Intel introduces the first microprocessor, designed by Ted Hoff. The 4004 had 4-bit buses and a clock rate of 108KHz. It had 2300 transistors and was built in a 10u process. It never captured much interest in the market and was soon eclipsed by its more capable brethren.



Exponential Growth



Introduced in 1972, the 8008 had 3,500 PMOS transistors supporting a byte-wide data path. Despite its limitations, the 8008 was the first microprocessor capable of playing the role of computer CPU as demonstrated on the cover of the July '74 issue of *Radio-Electronics*.



Last, but not least, on our tour is the 8080. Introduced in 1974, the 8080 had 6,000 NMOS transistors fab'ed in a 6u process. The clock rate was 2Mhz, more than enough to ignite the personal computer industry. At least Paul Allen and his partner thought so when they wrote a BASIC interpreter for the 8080 in 1975. They would later collaborate in another, more profitable, venture...



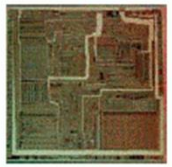
- http://www.intel.com/intel/intelis/museum/exhibit/hist_micro/hof/hof_main.htm



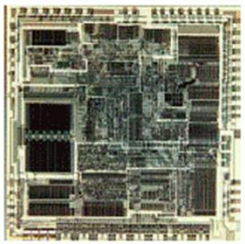
Shown with approximate relative sizes



Intel 8080A, 1974
3Mhz, 6K transistors, 6u



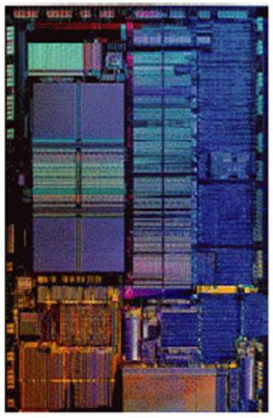
Intel 8086, 1978, 33mm^2
10Mhz, 29K transistors, 3u



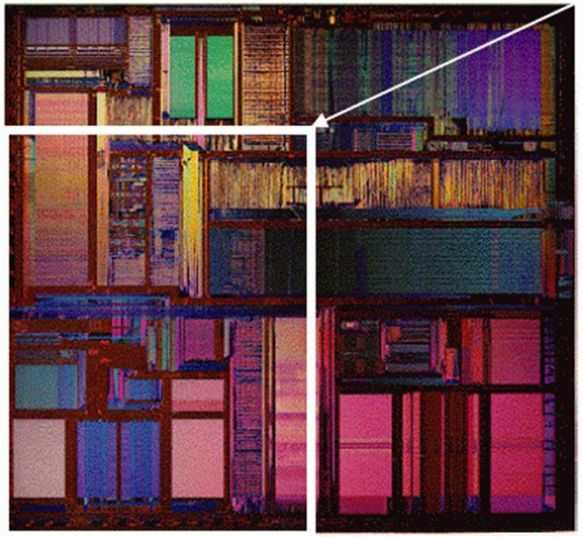
Intel 80286, 1982, 47mm^2
12.5Mhz, 134K transistors, 1.5u



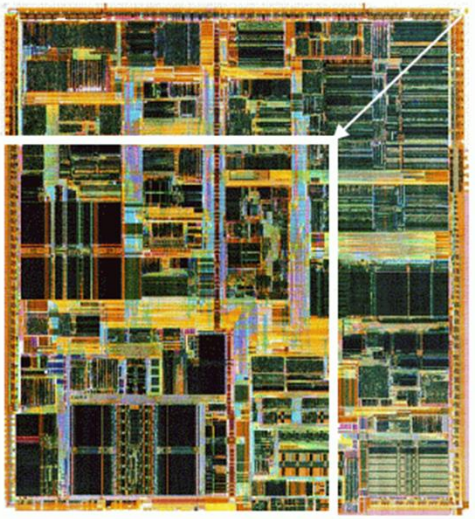
Intel 386DX, 1985, 43mm^2
33Mhz, 275K transistors, 1u



Intel 486, 1989, 81mm^2
50Mhz, 1.2M transistors, .8u



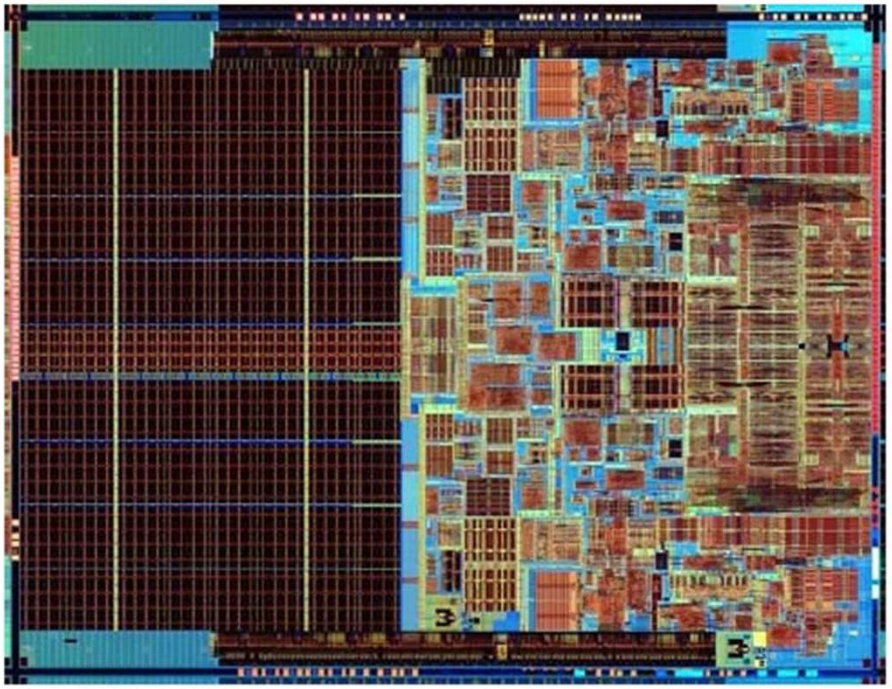
Intel Pentium, 1993/1994/1996, 295/147/90mm^2
66Mhz, 3.1M transistors, .8u/.6u/.35u



Intel Pentium II, 1997, 203mm^2/104mm^2
300/333Mhz, 7.5M transistors, .35u/.25u



Core 2 Duo: 291M Transistors (2006)

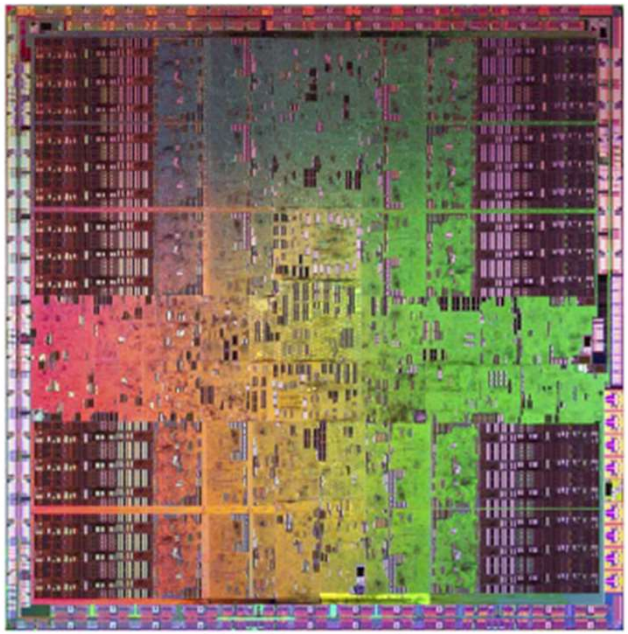


Core 2 Duo T5000/T7000 series mobile processors, called Penryn uses 800M of 45 nanometer devices (2007)

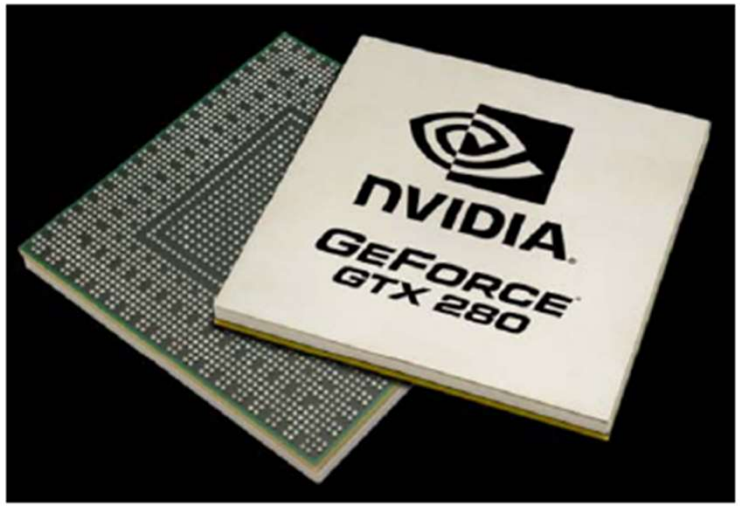


Nvidia GT200 Graphics Processing Unit

- This GPU contains 1.4 billion transistors.
- This uses a 65 nm CMOS fabrication process.



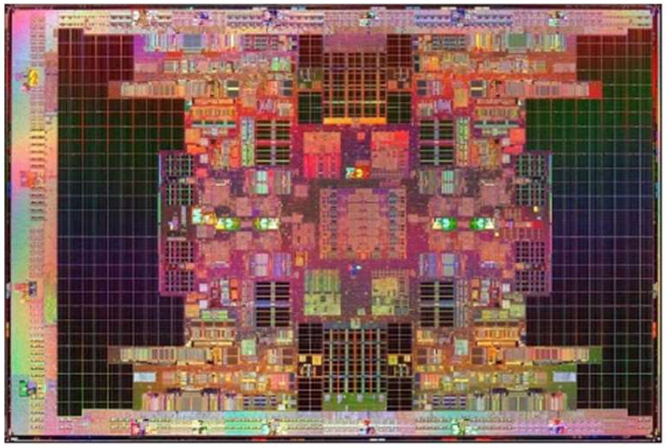
Source: http://www.gdiamos.net



Source:
http://nvidiathepoweroffuture.files.wordpress.com/



Intel's Tukwila – 2B Transistors



- Quad core i.e. 4 cores per die.
- 30 MB of cache.
- 2 billion transistors on the die.
- 64-bit Itanium family.
- 65nm CMOS technology.
- Release in early 2010.



Circuit Design Flow



Integrated Circuits Categories

- There are many different types of ICs as listed below.

IC Categories	Functions
Analog ICs	Amplifiers
	Filters
Digital ICs	Boolean Gates
	Encoders/Decoders
	Multiplexers / Demultiplexers
	Flip-flops
	Counters
	Shift Registers
Hybrid ICs	Mixed Signal Processors
Interface ICs	Analog-Digital Converters
	Digital-Analog Converters



Levels of Integration (Chip Complexity)

- Categorized by the number of gates contained in the chip.

IC Complexity	Number of Gates	Functional Complexity	Examples
SSI	<10	Basic gates	Inverters, AND gates, OR gates, NAND gates, NOR gates
MSI	10-100	Basic gates	Exclusive OR/NOR
		Sub-modules	Adders, subtractors, encoders, decoders, multiplexers, demultiplexers, counters, flip-flops
LSI	100-1000s	Functional modules	Shift registers, stacks
VLSI	1000s-100,000	Major building blocks	Microprocessors, memories
ULSI	>100,000	Complete systems	Single chip computers, digital signal processors
WSI	>10,000,000	Distributed systems	Microprocessor systems



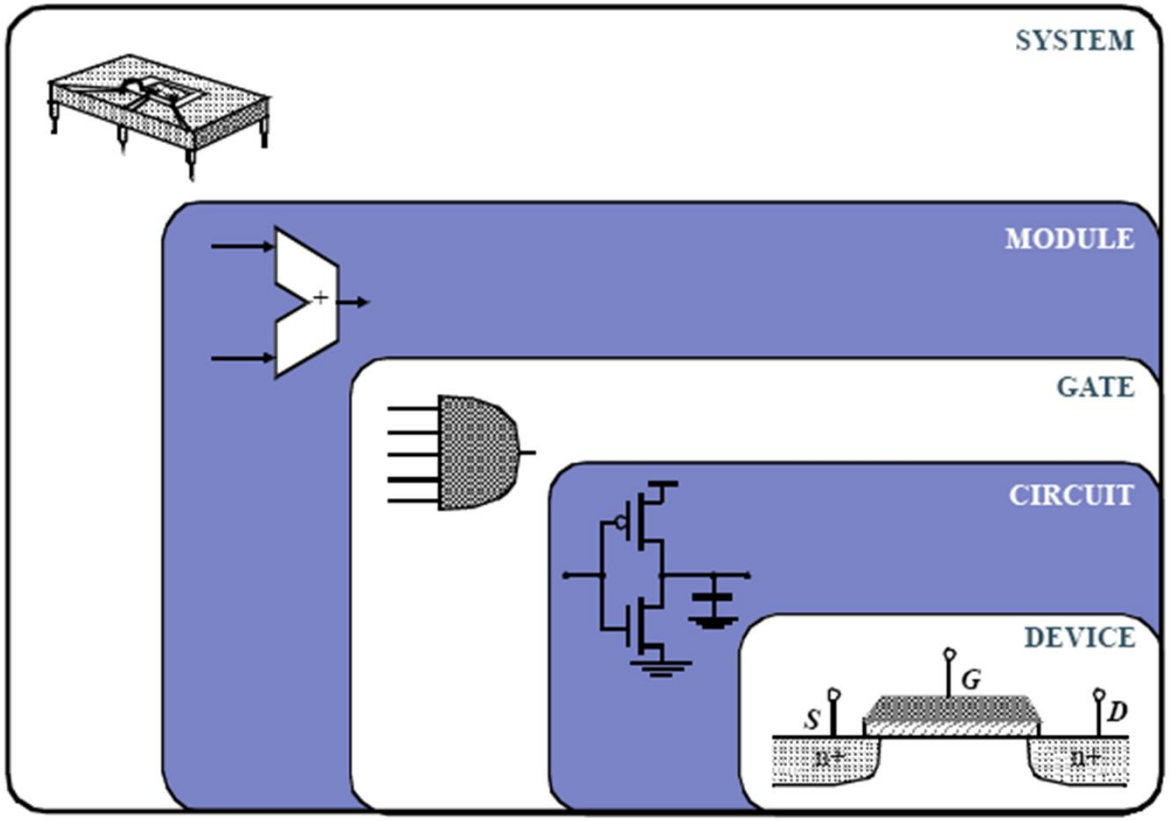
Digital Logic Families

- Various circuit technology used to implement an IC at lower level of abstraction.
- The circuit technology is referred to as a digital logic family.

RTL - Resistor-transistor Logic	obsolete
DTL - Diode-transistor logic	obsolete
TTL - Transistor-transistor logic	not much used
ECL - Emitter-coupled logic	high-speed ICs
MOS - Metal-oxide semiconductor	high-component density
CMOS - Complementary Metal-oxide semiconductor	widely used, low-power high-performance and high-packing density IC
BiCMOS - Bipolar Complementary Metal-oxide semiconductor	high current and high-speed
GaAs - Gallium-Arsenide	very high speed circuits

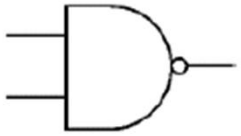


Design Abstraction Levels

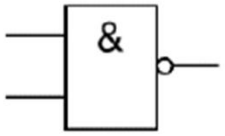




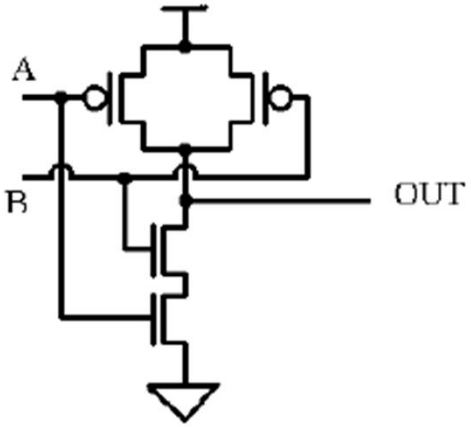
Digital Circuits : Logic to Device



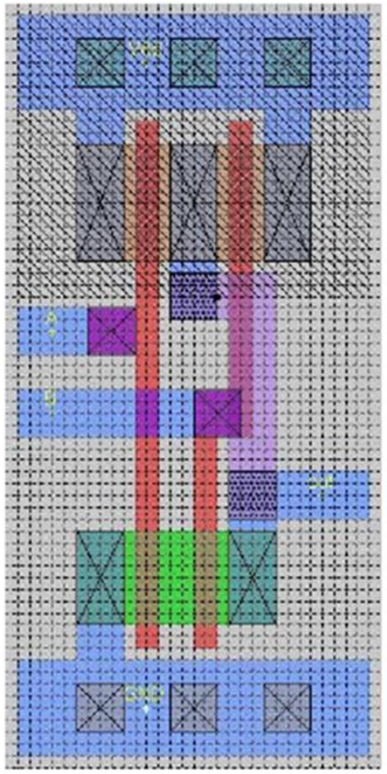
(NAND Gate)



(IEC Symbol)



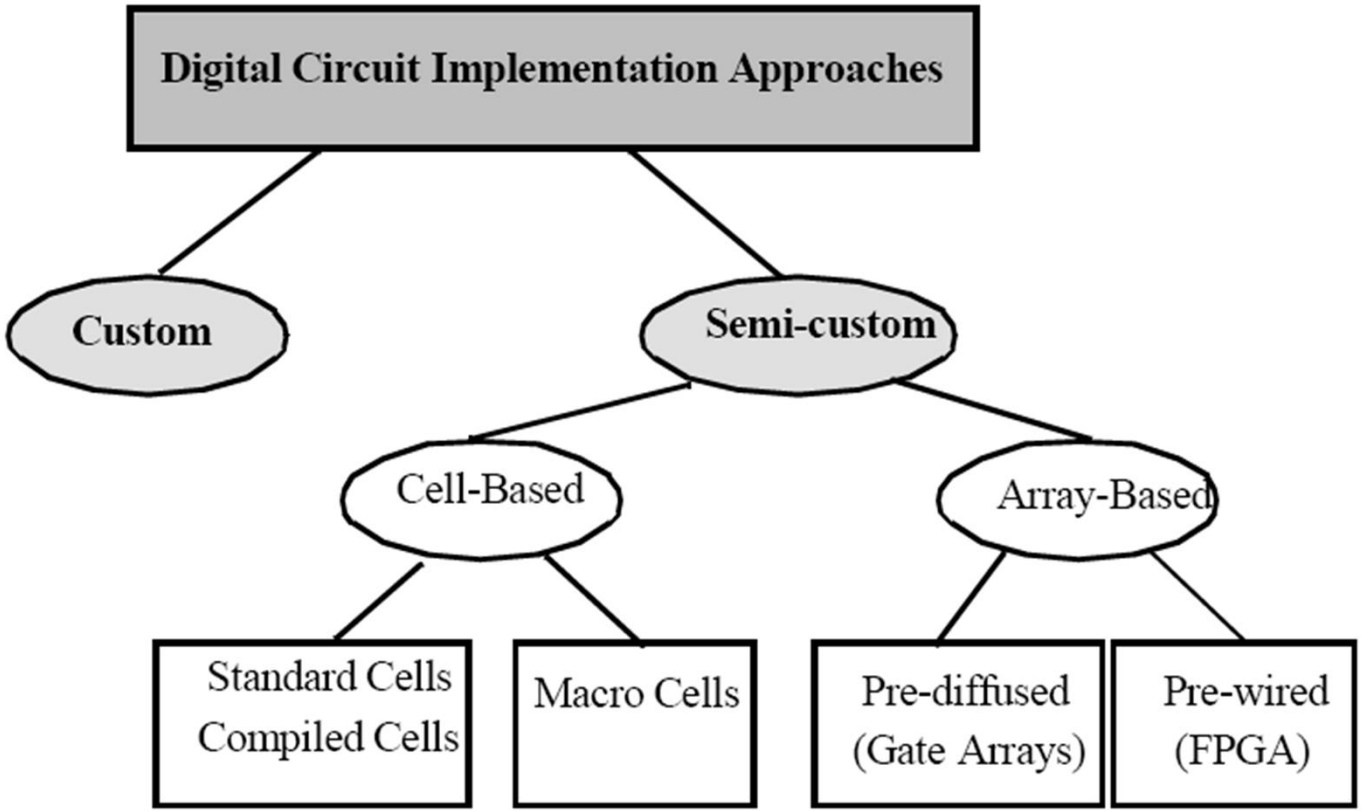
(Transistor Diagram)



(Layout Diagram)



Implementation Approaches for Digital ICs



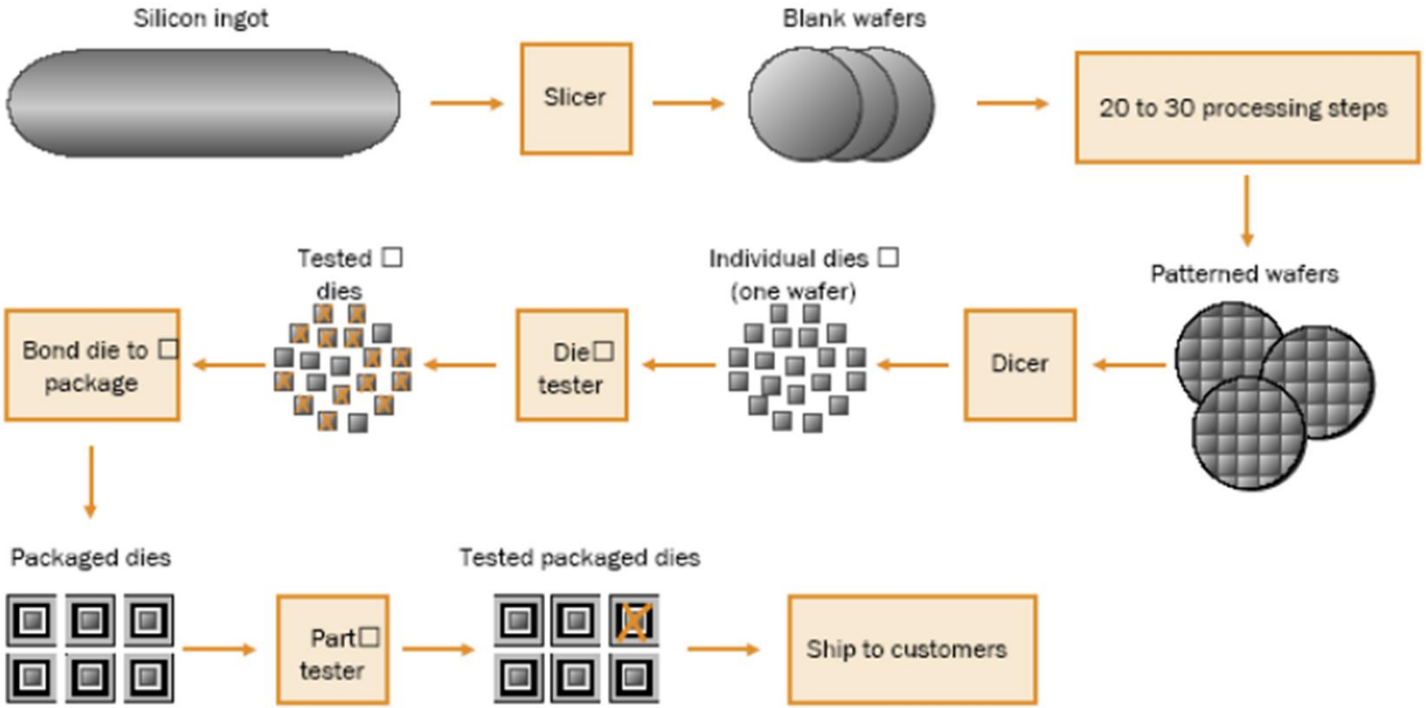


Implementation Approaches for Digital ICs

- Full-custom: all logic cells are customized. A general purpose microprocessor is designed this way.
- Semi-custom: all of the logic cells are from predesigned cell libraries (reduces the manufacture lead time of the IC)
- Standard-cell based IC uses predesigned logic cells Standard such as AND gates, OR gates, MUXs, FFs,.., etc.
- Macrocells (also called mega cells) are larger predesigned cells, such as microcontrollers, even microprocessors, etc.
- Gate-Array, Sea-of-Gates or prediffused arrays contains array of transistors or gates which can be connected by wires to implement the chip.
- Programmable-Logic-Array (PLA) is an example of fuse-based FPGA design. (NOTE: Fuse-based, nonvolatile and volatile are three types of FPGAs)



Digital IC Fabrication Flow





Technology Growth and Moore's Law



Different Attributes of an IC or chip

We will briefly discuss the VLSI technological growth based on these attributes.

- Transistor count of a chip
- Operating frequency of a chip
- Power consumption of a chip
- Power density in a chip
- Size of a device used in chip

NOTE: Chip is informal name for IC

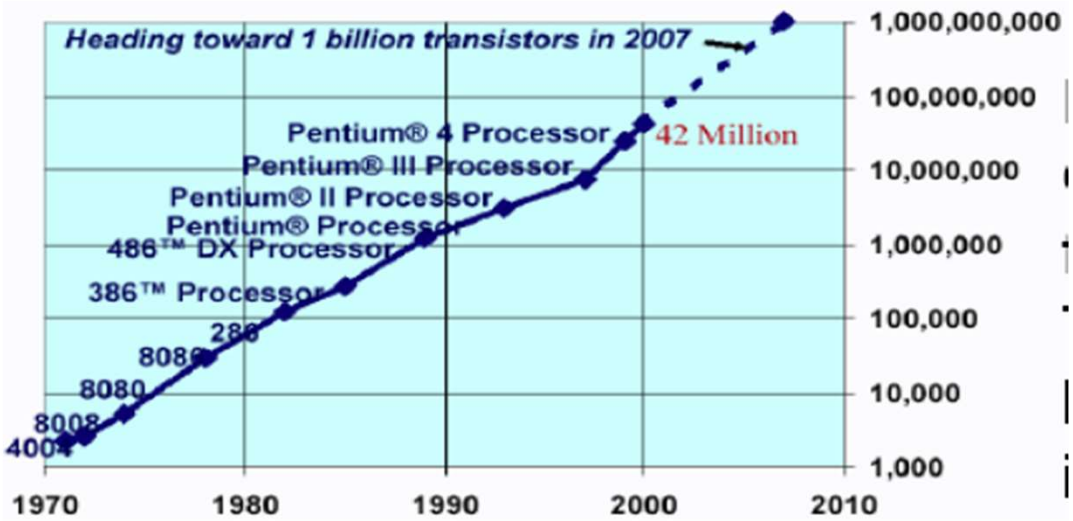


Moore's Law

- **1965: Gordon Moore plotted transistor on each chip**
  - Transistor counts have doubled every 26 months
- **Many other factors grow exponentially**
  - clock frequency
  - processor performance



VLSI Trend



Power, Operating frequency, and Throughput have increased.

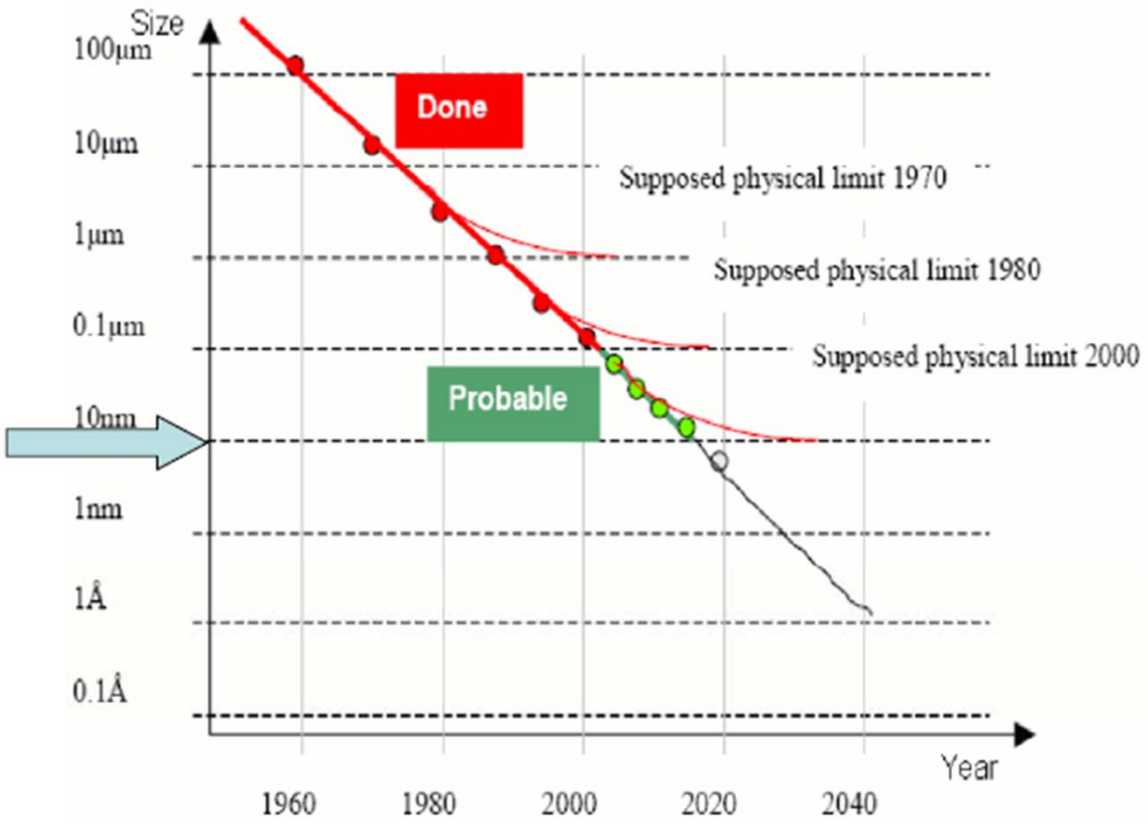


VLSI technology is the fastest growing technology in human history.

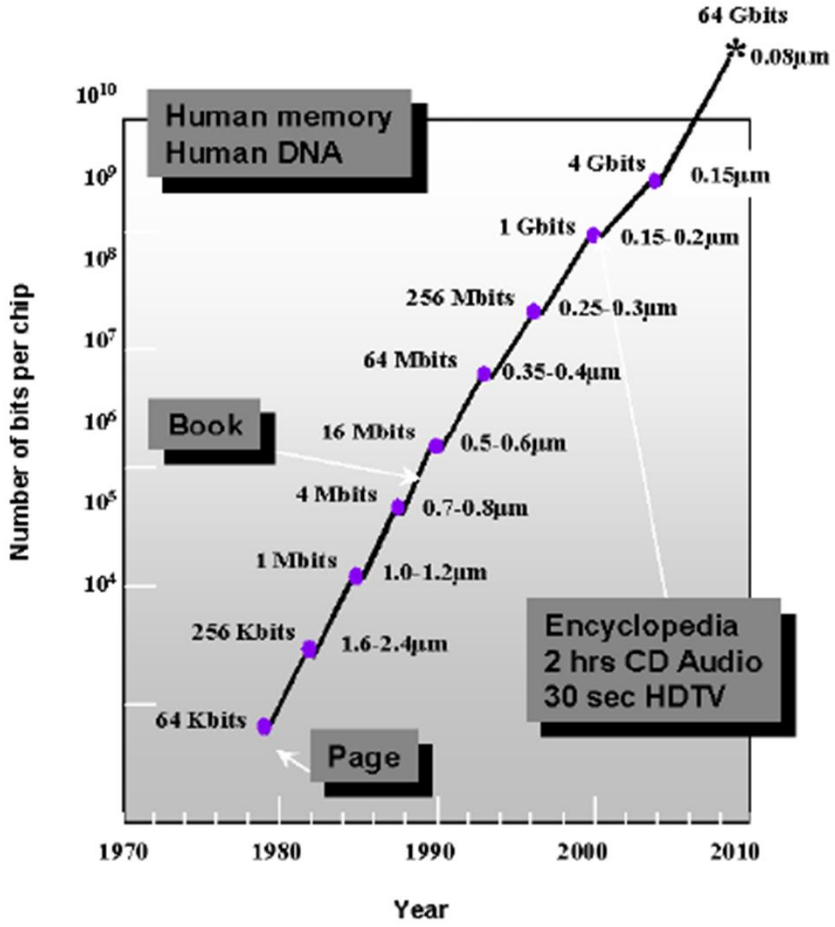




Technology Scaling Trend



Evolution in Complexity



Why Scaling?

- Technology shrinks by 0.7/generation
- With every generation can integrate 2x more functions per chip; chip cost does not increase significantly
- Cost of a function decreases by 2x
- But…

– How to design chips with more and more functions?

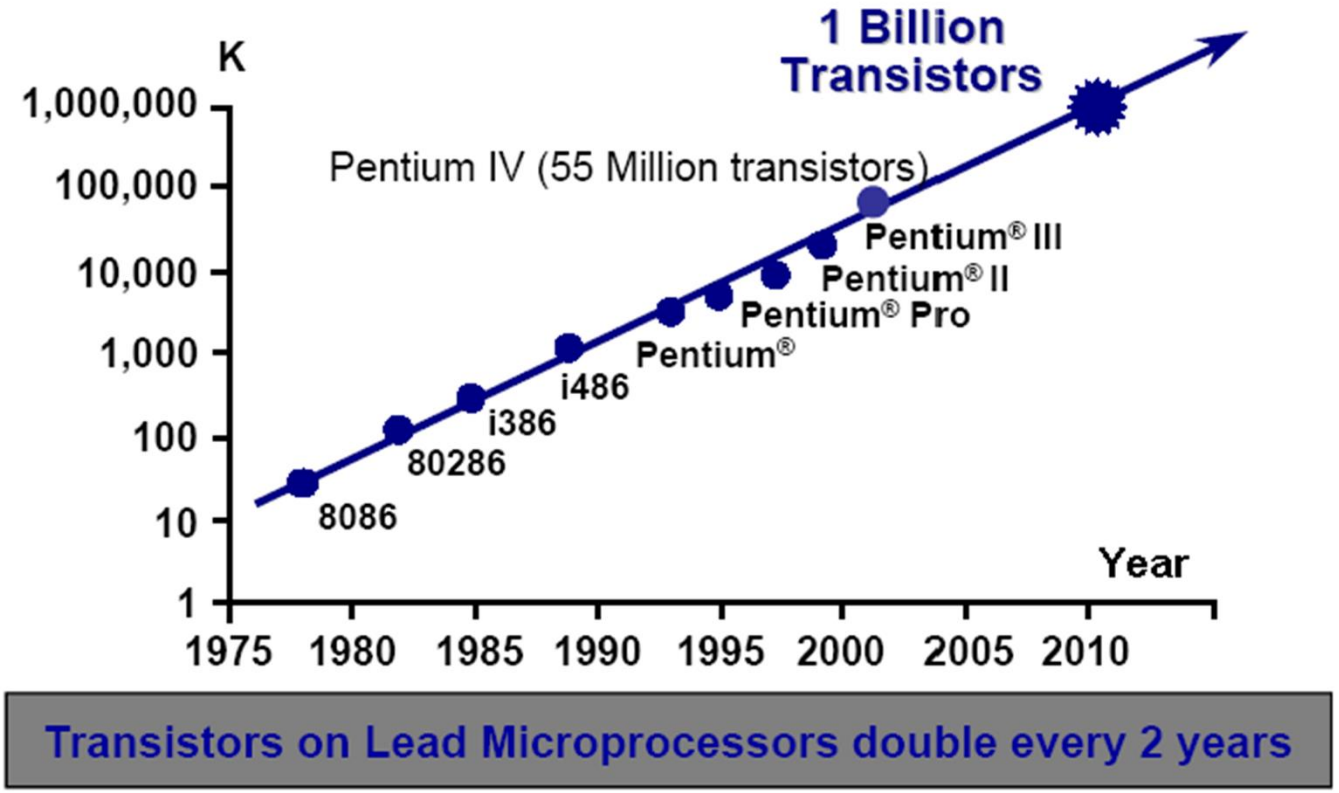
– Design engineering population does not double every two years…

- Hence, a need for more efficient design methods

– Exploit different levels of abstraction



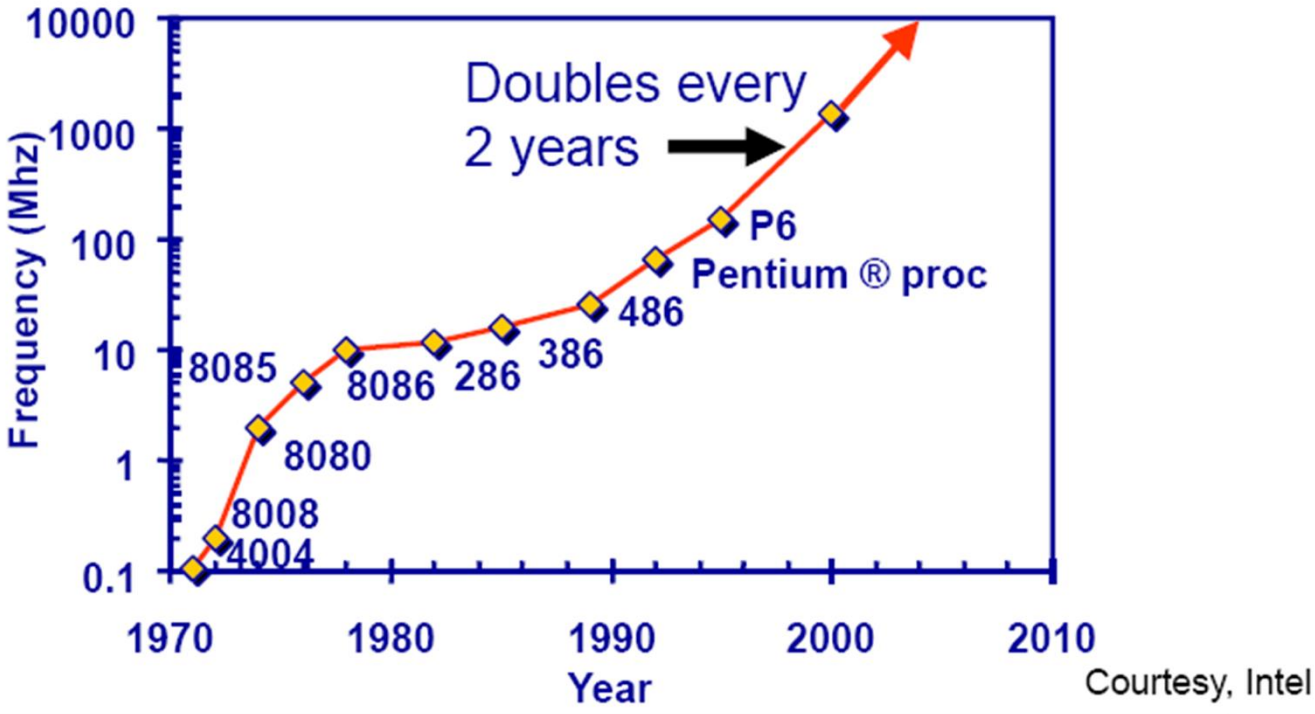
Increase in Transistor Count



Transistors on Lead Microprocessors double every 2 years



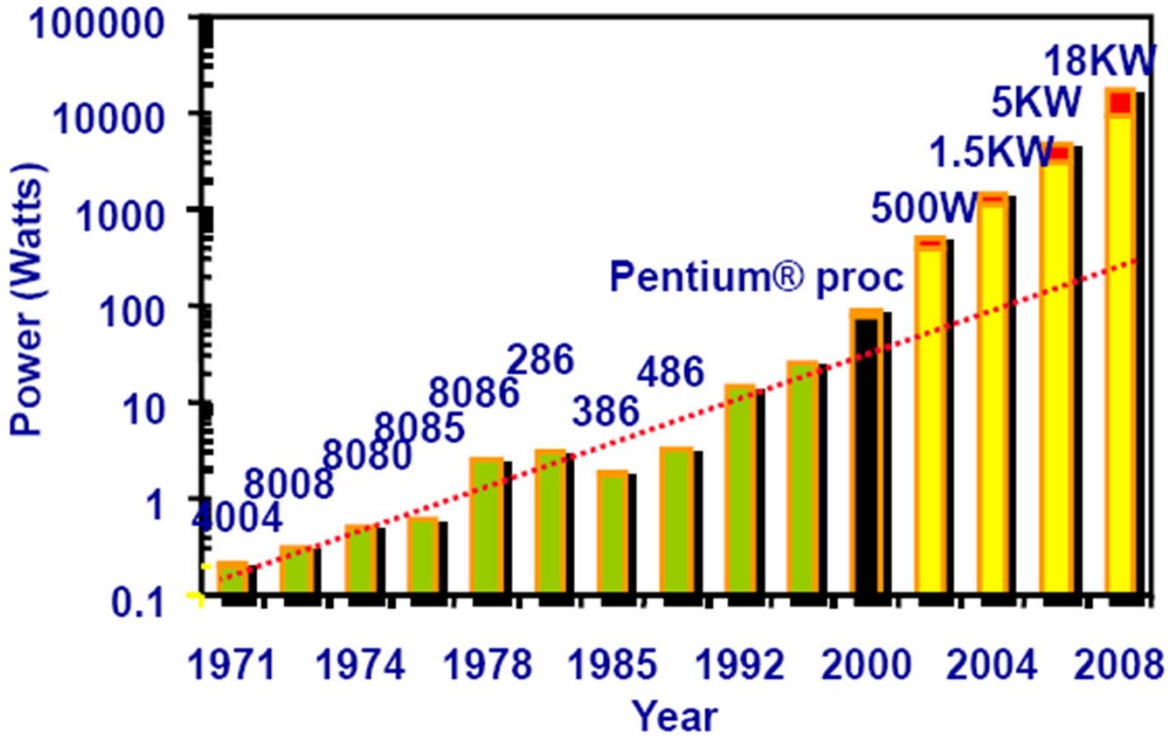
Increase in Operating Frequency



Lead Microprocessors frequency doubles every 2 years



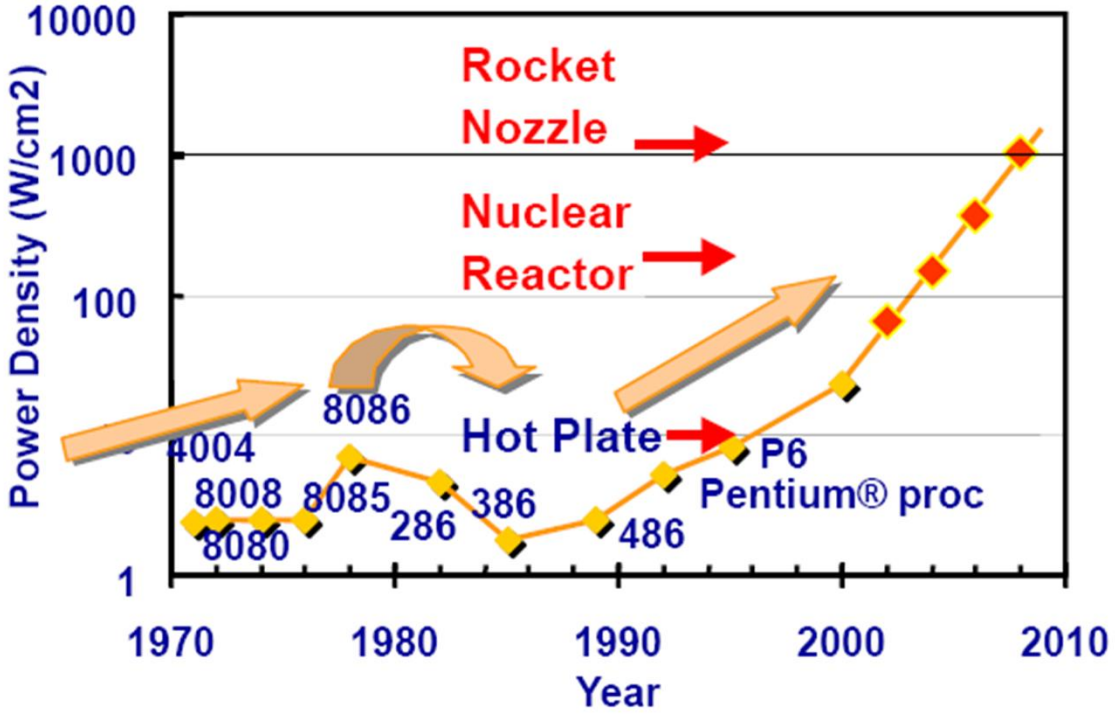
Power will be a major problem



Power delivery and dissipation will be prohibitive



Power density



Power density too high to keep junctions at low temp



Challenges in Digital Design

"Microscopic Problems"

- Ultra-high speed design
- Interconnect
- Noise, Crosstalk
- Reliability, Manufacturability
- Power Dissipation
- Clock distribution.

Everything Looks a Little Different



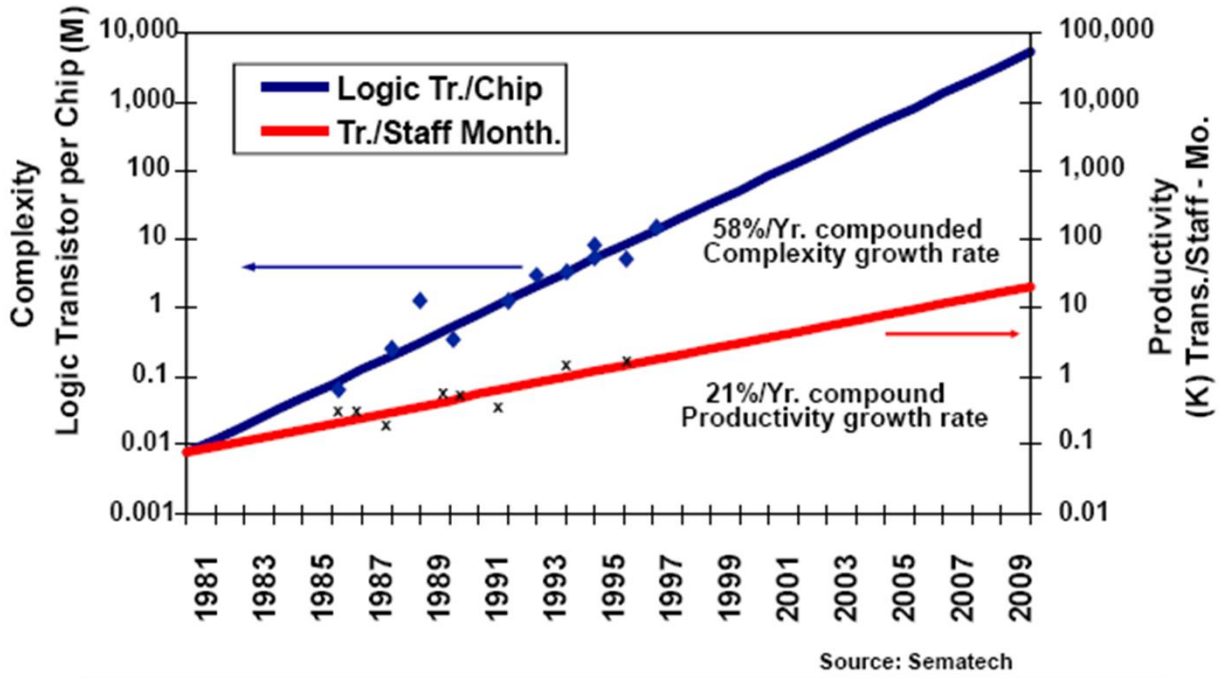
"Macroscopic Issues"

- Time-to-Market
- Millions of Gates
- High-Level Abstractions
- Reuse & IP: Portability
- Predictability
- etc.

…and There's a Lot of Them!



Productivity Trends



Complexity outpaces design productivity

Courtesy, ITRS Roadmap



ITRS - International Technology Roadmap for Semiconductors

YEAR	2002	2005	2008	2011	2014
TECHNOLOGY	130 nm	100 nm	70 nm	50 nm	35 nm
CHIP SIZE	400 mm^2	600 mm^2	750 mm^2	800 mm^2	900 mm^2
NUMBER OF TRANSISTORS (LOGIC)	400 M	1 Billion	3 Billion	6 Billion	16 Billion
DRAM CAPACITY	2 Gbits	10 Gbits	25 Gbits	70 Gbits	200 Gbits
MAXIMUM CLOCK FREQUENCY	1.6 GHz	2.0 GHz	2.5 GHz	3.0 GHz	3.5 GHz
MINIMUM SUPPLY VOLTAGE	1.5 V	1.2 V	0.9 V	0.6 V	0.6 V
MAXIMUM POWER DISSIPATION	130 W	160 W	170 W	175 W	180 W
MAXIMUM NUMBER OF I/O PINS	2500	4000	4500	5500	6000

Predictions of the worldwide semiconductor / IC industry about its own future prospects...



Shrinking Device Dimensions

YEAR	2002	2005	2008	2011	2014
TECHNOLOGY	130 nm	100 nm	70 nm	50 nm	35 nm
CHIP SIZE	400 mm^2	600 mm^2	750 mm^2	800 mm^2	900 mm^2
NUMBER OF TRANSISTORS (LOGIC)	400 M	1 Billion	3 Billion	6 Billion	16 Billion
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MAXIMUM NUMBER OF I/O PINS	2500	4000	4500	5500	6000



Increasing Function Density

YEAR	2002	2005	2008	2011	2014
TECHNOLOGY	130 nm	100 nm	70 nm	50 nm	35 nm
CHIP SIZE	400 mm^2	600 mm^2	750 mm^2	800 mm^2	900 mm^2
NUMBER OF TRANSISTORS (LOGIC)	**400 M**	**1 Billion**	**3 Billion**	**6 Billion**	**16 Billion**
DRAM CAPACITY	**2 Gbits**	**10 Gbits**	**25 Gbits**	**70 Gbits**	**200 Gbits**
MAXIMUM CLOCK FREQUENCY	1.6 GHz	2.0 GHz	2.5 GHz	3.0 GHz	3.5 GHz
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MAXIMUM POWER DISSIPATION	130 W	160 W	170 W	175 W	180 W
MAXIMUM NUMBER OF I/O PINS	2500	4000	4500	5500	6000



Increasing Clock Frequency

YEAR	2002	2005	2008	2011	2014
TECHNOLOGY	130 nm	100 nm	70 nm	50 nm	35 nm
CHIP SIZE	400 mm^2	600 mm^2	750 mm^2	800 mm^2	900 mm^2
NUMBER OF TRANSISTORS (LOGIC)	400 M	1 Billion	3 Billion	6 Billion	16 Billion
DRAM CAPACITY	2 Gbits	10 Gbits	25 Gbits	70 Gbits	200 Gbits
MAXIMUM CLOCK FREQUENCY	**1.6 GHz**	**2.0 GHz**	**2.5 GHz**	**3.0 GHz**	**3.5 GHz**
MINIMUM SUPPLY VOLTAGE	1.5 V	1.2 V	0.9 V	0.6 V	0.6 V
MAXIMUM POWER DISSIPATION	130 W	160 W	170 W	175 W	180 W
MAXIMUM NUMBER OF I/O PINS	2500	4000	4500	5500	6000



Decreasing Supply Voltage

YEAR	2002	2005	2008	2011	2014
TECHNOLOGY	130 nm	100 nm	70 nm	50 nm	35 nm
CHIP SIZE	400 mm^2	600 mm^2	750 mm^2	800 mm^2	900 mm^2
NUMBER OF TRANSISTORS (LOGIC)	400 M	1 Billion	3 Billion	6 Billion	16 Billion
DRAM CAPACITY	2 Gbits	10 Gbits	25 Gbits	70 Gbits	200 Gbits
MAXIMUM CLOCK FREQUENCY	1.6 GHz	2.0 GHz	2.5 GHz	3.0 GHz	3.5 GHz
MINIMUM SUPPLY VOLTAGE	**1.5 V**	**1.2 V**	**0.9 V**	**0.6 V**	**0.6 V**
MAXIMUM POWER DISSIPATION	**130 W**	**160 W**	**170 W**	**175 W**	**180 W**
MAXIMUM NUMBER OF I/O PINS	2500	4000	4500	5500	6000



Circuit Design Metrics



VLSI Design

This is a good news/bad news story. Although the cost of manufacturing IC's remained constant, the design cost did not. In fact, it grew exponentially with the complexity of the circuit. The integrating of a system on a piece of silicon had an attractive manufacturing cost, but frightening was (and is) fairly limited.

This course is an attempt to help solve some of these design problems. By extracting out the basic features of integrated system design, it is possible to teach a much wider class of people about integrated circuits; by using a structured design style, some of the long design times can be circumvented.



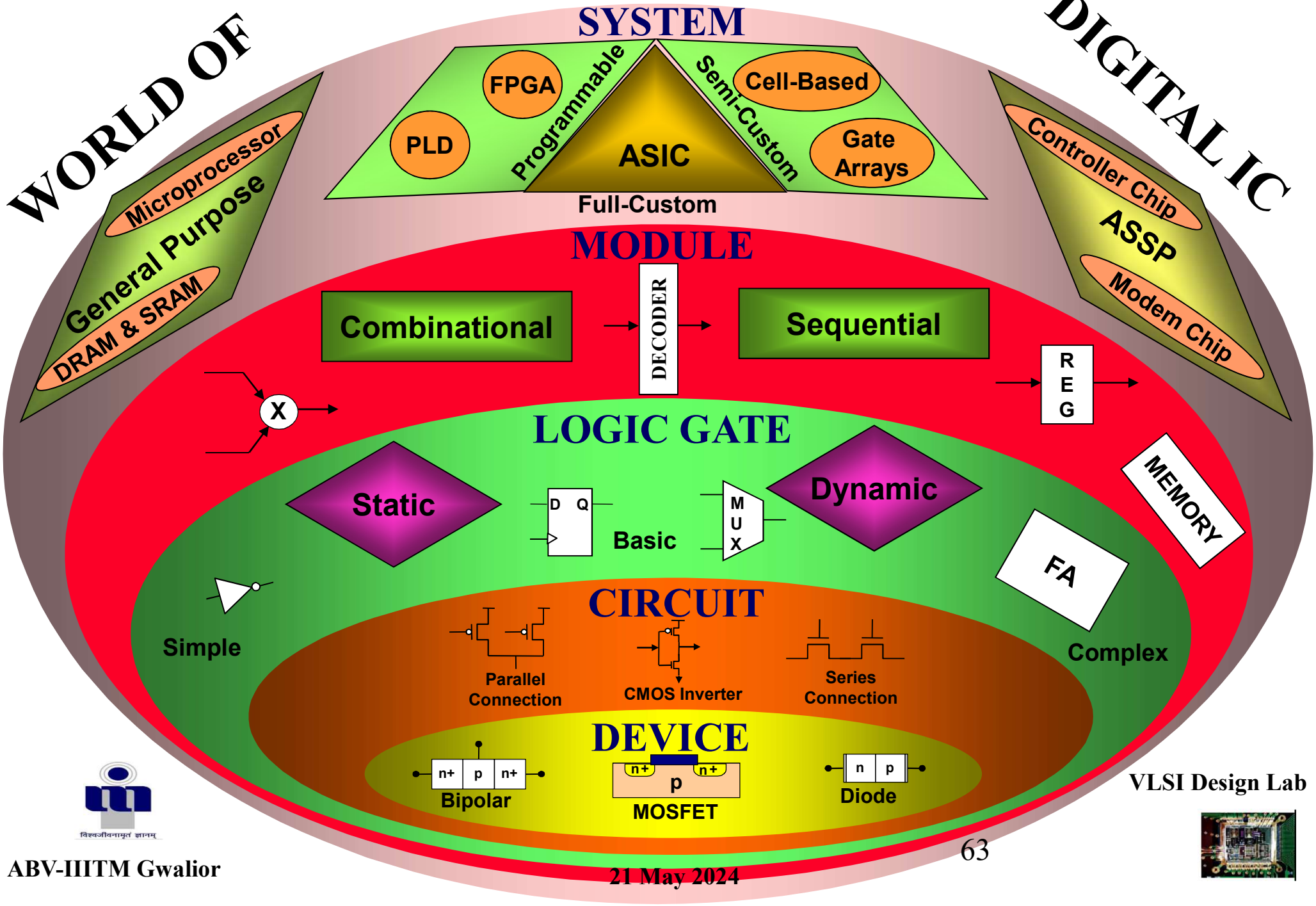
Profound Impact on the way VLSI is Designed

- The old way: manual transistor twiddling
  - expert "layout designers"
  - entire chip hand-crafted
  - okay for small chips… but cannot design billion transistor chips in this fashion

- The new way: using ECAD tools at high level
  - tools do the grunge work…
  - high levels of abstractions
    synthesis from a description of the behavior
  - libraries of reusable cores, modules, and cells

Chip design increasingly like object-oriented software design!




WORLD OF
DIGITAL IC
SYSTEM
Programmable
Semi-Custom
FPGA
PLD
ASIC
Cell-Based
Gate Arrays
Full-Custom
Microprocessor
General Purpose
DRAM & SRAM
Controller Chip
ASSP
Modem Chip
MODULE
Combinational
DECODER
Sequential
REG
X
LOGIC GATE
Static
Dynamic
D Q
Basic
MUX
MEMORY
FA
Simple
Complex
CIRCUIT
Parallel Connection
CMOS Inverter
Series Connection
DEVICE
n+ p n+
Bipolar
n+ p n+
p
MOSFET
n p
Diode

MERITS IN VLSI CIRCUITS AND DESIGN

- FUNCTIONALITY
- PERFORMANCE
- RELIABILITY
- ENERGY CONSUMPTION
- AREA (PACKING DENSITY)



- FUNCTIONALITY: Ability to produce proper outputs from given inputs
- PERFORMANCE: High speed, low delay
- RELIABILITY: Robustness and noise tolerance in adverse conditions and temperatures
- ENERGY CONSUMPTION: Low power to allow portability and avoid costly cooling systems
- AREA: Amount of silicon used and processing cost.



VLSI Benefits

- Smaller Size
- Higher Performance
- Higher Functionality
- Higher Reliability
- Lower Power Consumption
- Design Security



VLSI System Design Challenges

1. Availability of advanced design tools
2. Design of system-on-chip
3. RF and Analog/Mixed Signals (RF-AMS) integration
4. Lithography and scaling
5. Design of programmable/reconfigurable architectures
6. Low-power/high-performance designs
7. Optoelectronics and integrated optics
8. Quantum and biological computing
9. Mechatronics, medical, and biological applications



Business Rules

- **Moore's Law:**
  Computer capacity in execution speed / memory capacity doubles every 18 month;

- **Gilder's Law:**
  Communication bandwidth doubles every 12 month;

- **Metcalf's Law:**
  Network value grows quadratic with the increase in users;

- **Arthur's Law:**
  Internet business has a growing return on investment (ROI).



Design Metrics

- How to evaluate performance of a digital circuit (gate, block, …)?

–Cost

– Reliability

– Scalability

– Speed (delay, operating frequency)

– Power dissipation

– Energy to perform a function



Full-custom ASIC

- All mask layers are customized in a **full-custom ASIC**.
- It only makes sense to design a full-custom IC if there are no libraries available.
- Full-custom offers the highest performance and lowest part cost (smallest die size) with the disadvantages of increased design time, complexity, design expense, and highest risk.
- Microprocessors were exclusively full-custom, but designers are increasingly turning to semicustom ASIC techniques in this area too.
- Other examples of full-custom ICs or ASICs are requirements for high voltage (automobile), analog/digital (communications), or sensors and actuators.
- Until Recently, Unthinkable Expensive Development
  - Risky, Special Skills, Lack of Manpower

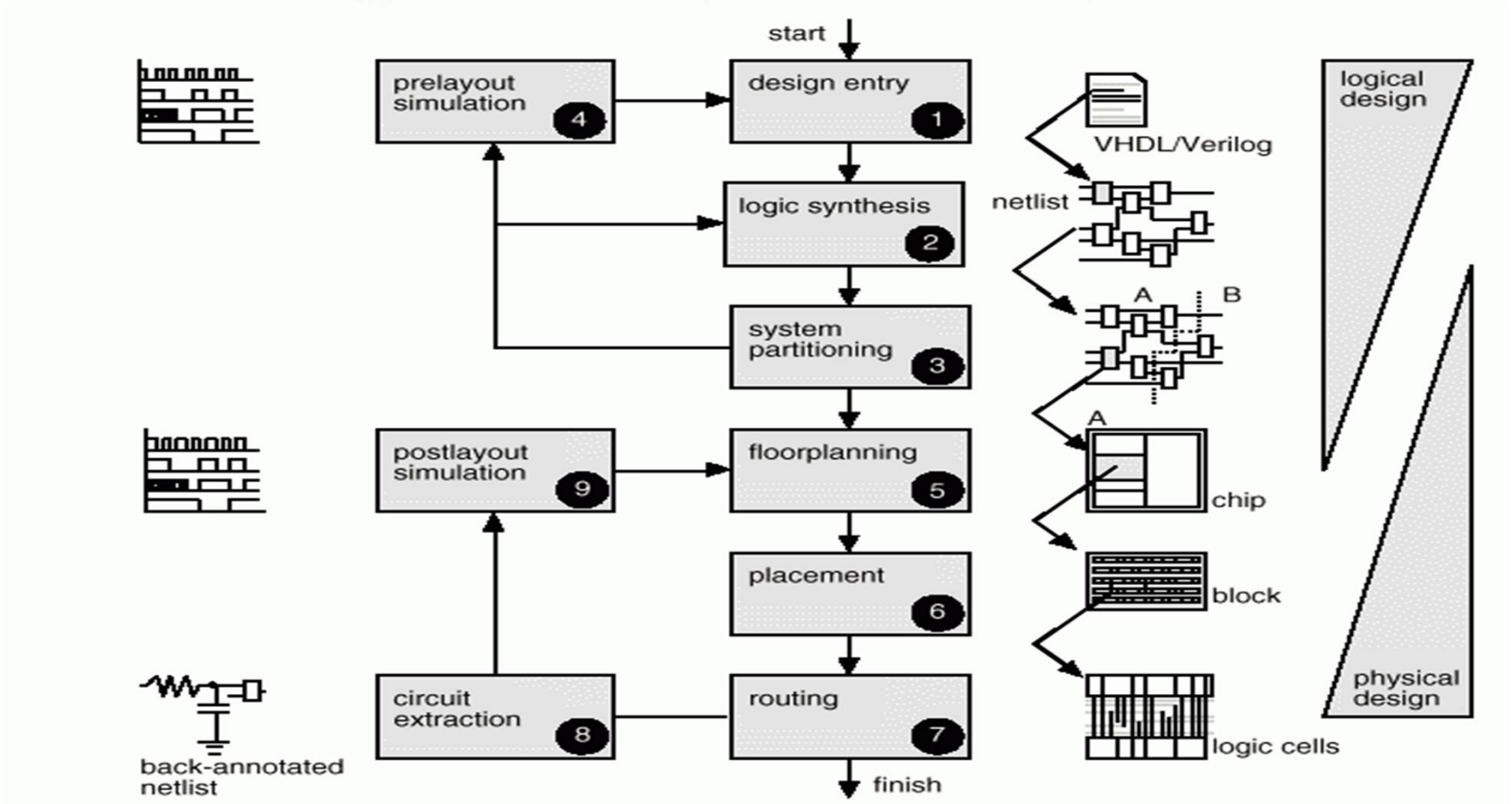


Design Flow

A design flow is a sequence of steps to design an ASIC

- **Design entry**. Using a **hardware description language** (**HDL**) or schematic entry.
- **Logic synthesis**. Produces a **netlist**—logic cells and their connections.
- **System partitioning**. Divide a large system into ASIC-sized pieces.
- **Pre-layout simulation**. Check to see if the design functions correctly.
- **Floorplanning**. Arrange the blocks of the netlist on the chip.
- **Placement**. Decide the locations of cells in a block.
- **Routing**. Make the connections between cells and blocks.
- **Extraction**. Determine the resistance and capacitance of the interconnect.
- **Postlayout simulation**. Check to see the design still works with the added loads of the interconnect.

Design Flow (continued)



ASIC design flow. Steps 1–4 are **logical design**, and steps 5–9 are **physical design**



ASIC Cell Libraries

- Use a **design kit** from the **ASIC vendor**
  - Usually a **library**—the cells are empty boxes, you **hand off** your design to the ASIC vendor and they perform box **instantiation** (Synopsys CBA)

- Buy an **ASIC-vendor library** from a **library vendor**
  - **Buy-or-build decision**. You need a **qualified cell library** (qualified by the **ASIC foundry**) If you own the masks (the **tooling**) you have a **customer owned tooling** solution (which is becoming very popular)

- Build your own cell library
  - Involves a complex **library development** process: **cell layout** ; **behavioral model** ; Verilog/VHDL model ; **timing model ; test strategy ;characterization ; circuit extraction ; process control monitors** (**PCMs**) or **drop-ins** ; **cell schematic** ; cell icon ; **layout versus schematic** (**LVS**) check; **cell icon** ; **logic synthesis** ; **retargeting** ; **wire-load model** ; **routing model;**



Electronic Design Automation (EDA)

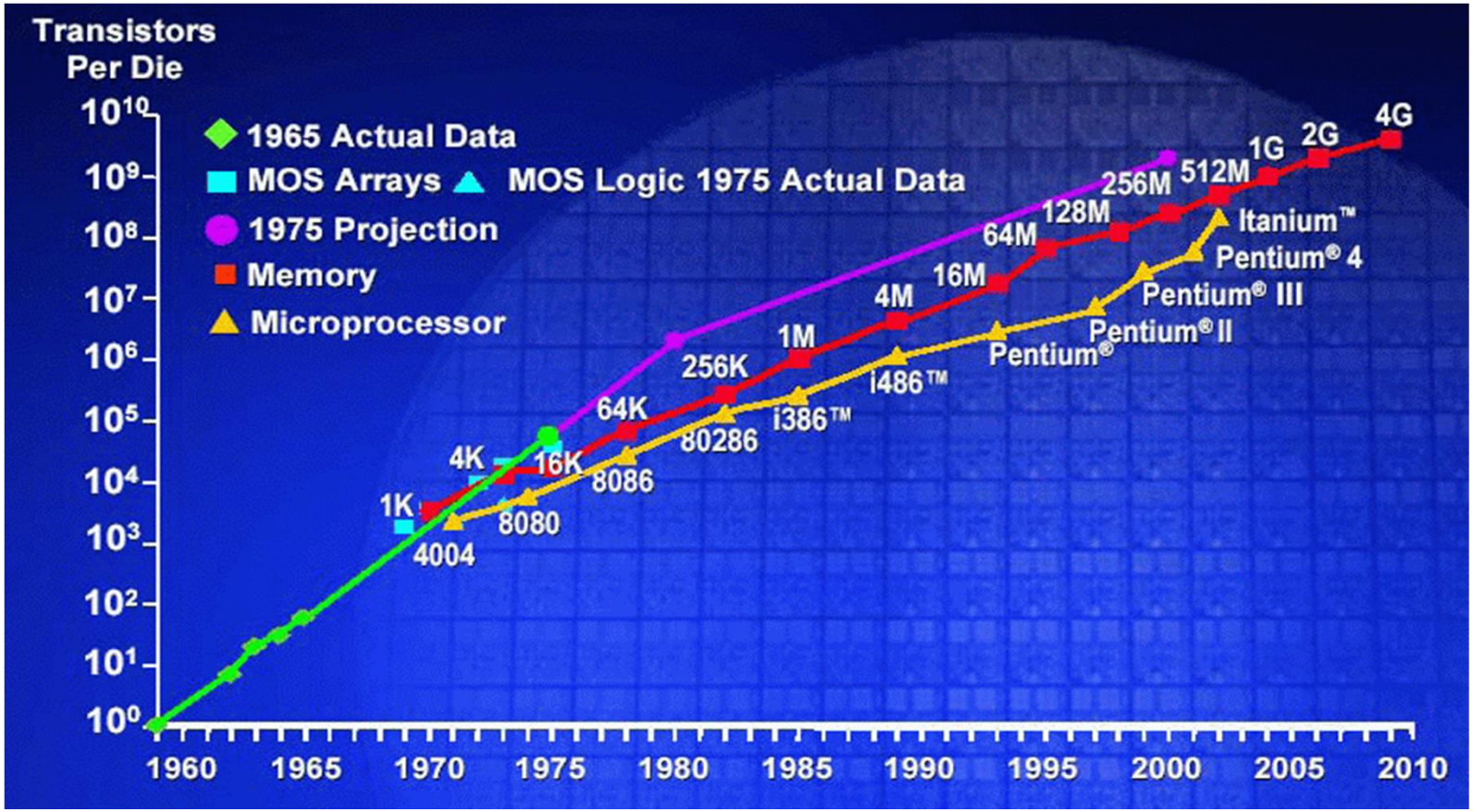
- **Computer-aided design (CAD):** is the use of a wide range of computer-based tools that assist engineers in their design activities. It involves both software and special-purpose hardware.

- **Electronic design automation (EDA):** is used to refer to the category of tools for designing and producing electronic systems ranging from printed circuit boards (PCBs) to ICs. This is sometimes referred to as ECAD (Electronic Computer Aided Design).

- EDA is used as an umbrella term for computer-aided engineering, computer-aided design, and computer-aided manufacturing of electronics in the discipline of electrical engineering.



Moore's Law

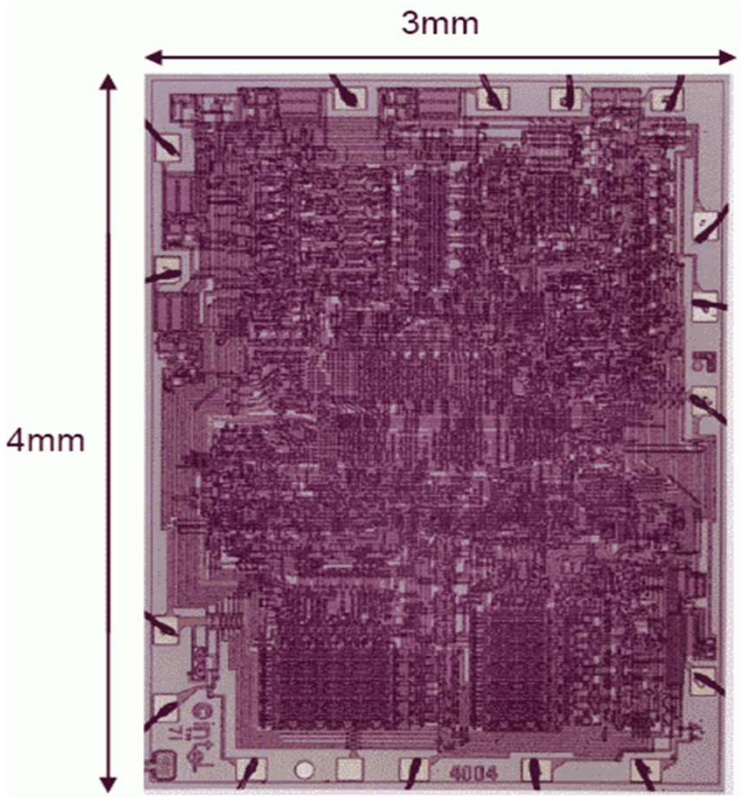


Original "law": number of components on IC doubles every year

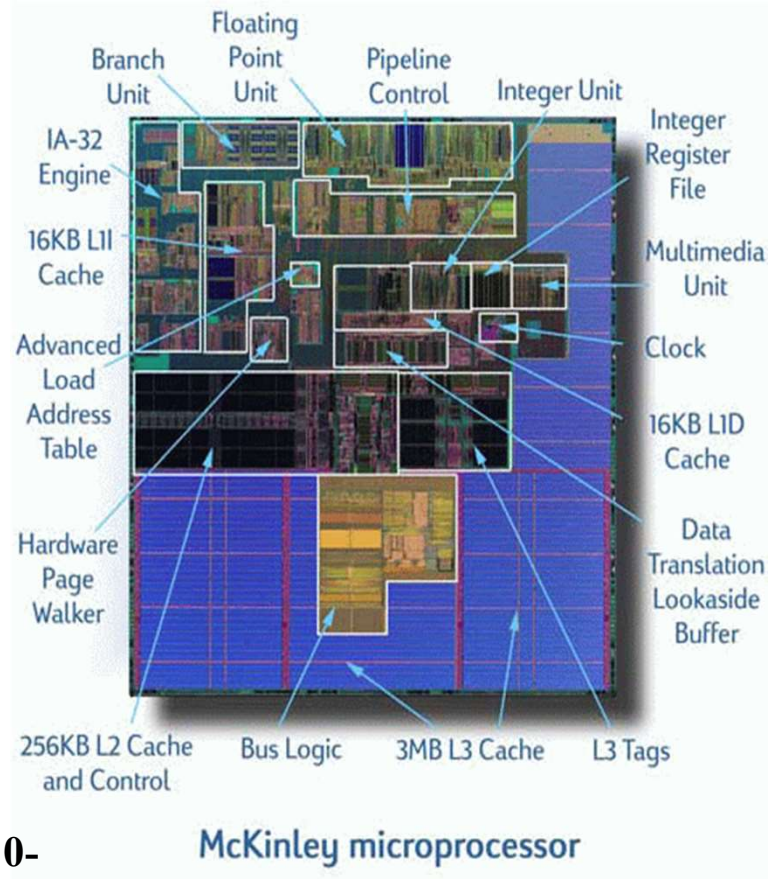
Later modified to doubling in every 18 months



Modern IC's



intel4004 4-bit µP('71, 2.3K transistors, 10-um technology, 108-kHz)



McKinley microprocessor

Itanium 2

2002

1-GHz

130-W

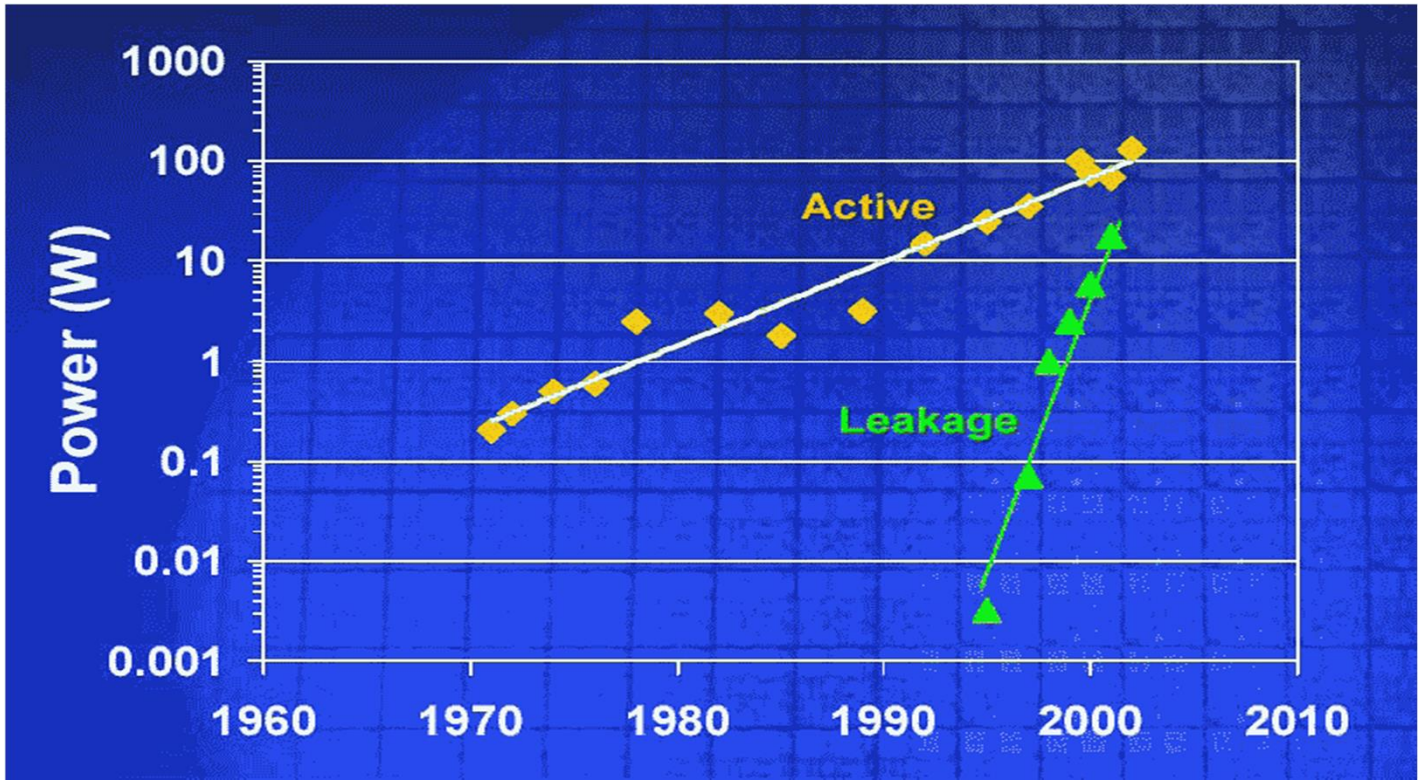
0.18-µm

221M transistors

421-mm^2(~20 x 21 mm)



But Performance Comes at a Cost



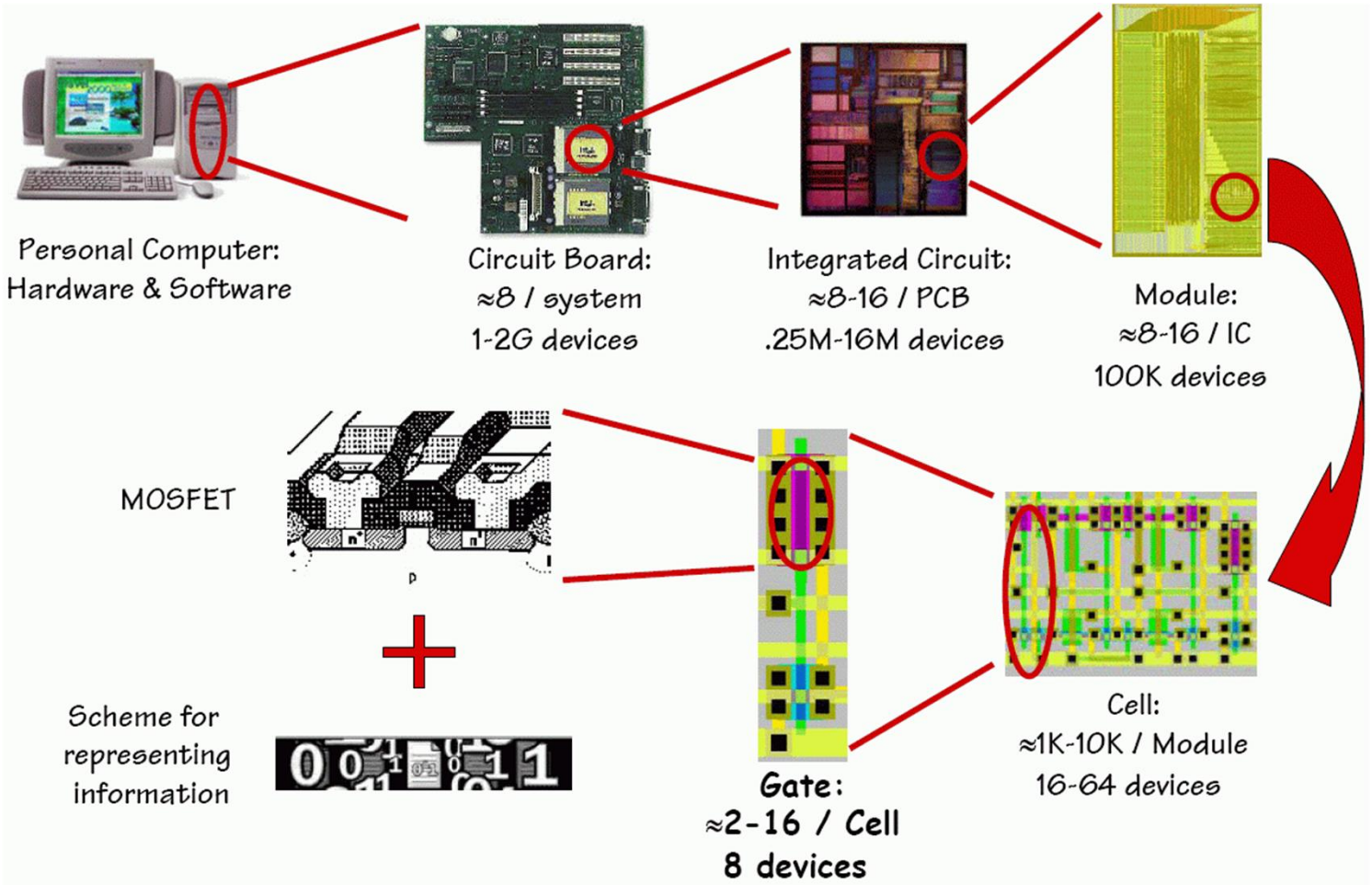
- Power is also becoming a problem…it's expensive (package, fans, etc.)
- The narrowing gap between active and leakage is also a growing problem
  - –Burns power even when the chip is not doing anything



- Things getting worse with each generation:
  - Complexity
  - Power consumption
  - Power distribution
  - Signal integrity
  - Fabrication time
  - Capital costs for equipment
  - Fixed costs for masks
  - Variable costs for wafers
- How do we handle these for a device with 100M gates?



Modern systems engineering



VLSI Design

Managing Complexity

- Simplify the design problem
  - Can't understand 10M transistors, or 100M rectangles
  - Need to make less complex (and less numerous) models
  - Abstraction
    - Simplified model for a thing, works well in some subset of the design space
  - Constraints
    - Needed to ensure that the abstractions are valid
    - Might work if you violate constraints, but guarantees are off
- Understand the underlying technology
  - Provide a feeling for what abstractions and constraints are needed
  - Determine efficient solutions (in design time, or implementation area, power, or performance)



Abstractions and Disciplines
How to Deal with 10 Transistors

- Digital abstraction
  - signals are 1 or 0
- Switch abstraction
  - MOSFETs as simple switches
- Gate abstraction
  - Separable timing
- Synchronous abstraction
  - Function does not depend on timing

- Constrain the design space to simplify the design process
  - strike a balance between design complexity and absolute performance

- Partition the problem(Use hierarchy)
  - Module is a box with pins
  - apply recursively



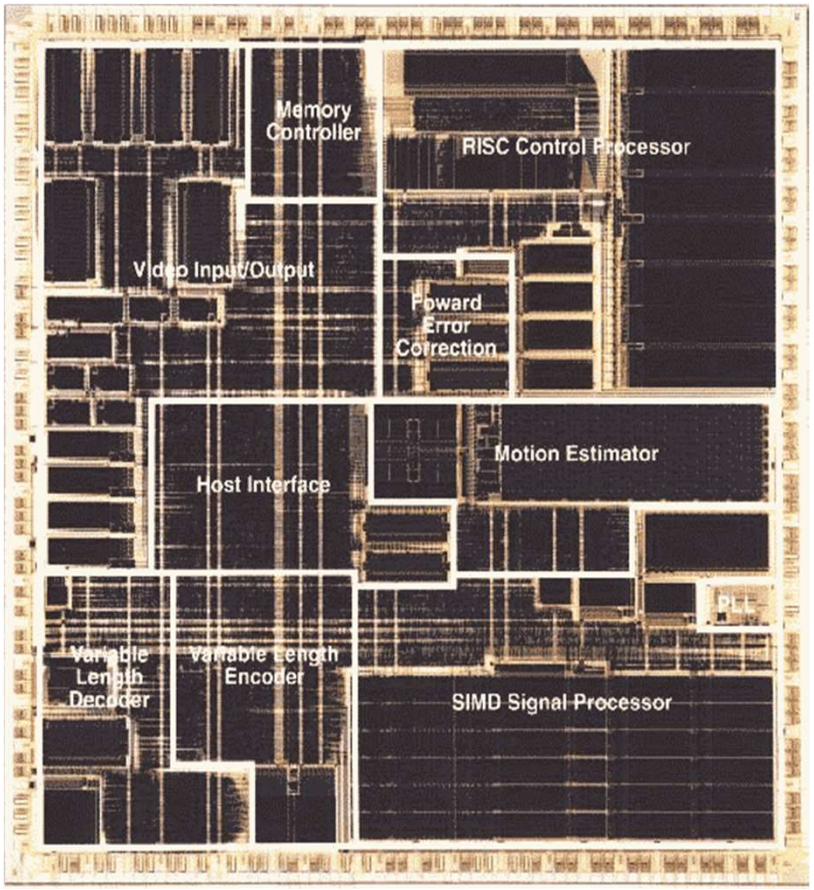
Design Levels

- **Specification**
  - what the system (or component) is supposed to do
- **Architecture**
  - high- level design of component
    - state defined
    - logic partitioned into major blocks
- **Logic**
  - gates, flip-flops, and the connections between them
- **Circuit**
  - transistor circuits to realize logic elements
- **Device**
  - behavior of individual circuit elements
- **Layout**
  - geometry used to define and connect circuit elements
- **Process**
  - steps used to define circuit elements

Can describe design at many different levels of abstraction



The Modern Era



AVP-III Video Codec from Lucent Technologies

- Multiple functional blocks (some general purpose) stitched together at the top level to make video encoder/decoder. Lots of modeling at architectural level to ensure that functional goals could be met.
- Different teams worked on the different blocks. Note that each block is itself composed of subblocks, and so on for many levels of hierarchy. Lots of iteration and reuse…
- Many different architectural choices: RISC controller, SIMD compute engine, special purpose motion estimator. Each choice made to meet some performance and area goal. Used lots of clever implementation tricks: the "obvious" implementation derived from the spec would require a chip many times this size (and hence impossible to build today).
- 2.5M transistors required lots of support from CAD tools for assembling the blocks, doing the routing and VERIFYING that everything was hooked up okay.



Architectural choices

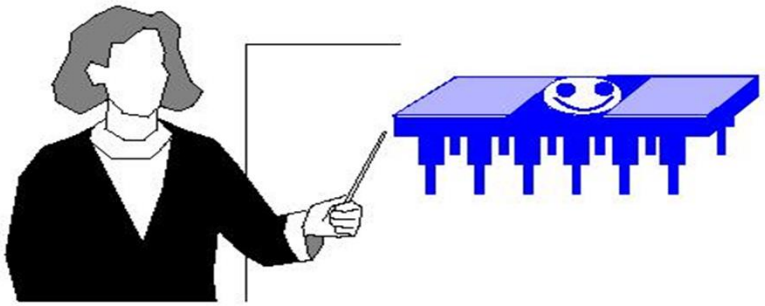
- Reducing marketing and budget input to an engineering specification that meets the constraints
- Deciding between general-purpose (programmable) vs. special-purpose (hard-wired) implementations for each major functional block.
- Choosing the appropriate architecture for each block trading off latency, throughput, area, power, design time, verifiability, wire densities, …
- Choosing the appropriate implementation strategy for each block



VLSI Design Process

The Design Process: An iterative process that refines an "idea" to a manufacturable device through at least five levels of design abstraction.

Top level: The "idea" → refined into a set of requirements called **Specification:**



- What does the chip do?
- How fast does it need to operate in order to be competitive?
- How much power will it consume?
- How big will it be?

Design Constraints:
Speed, power and area.

Abstraction: A very effective means of dealing with design complexity. Creating a model at a higher level of abstraction involves replacing detail at the lower level with simplifications.



Implementation metrics

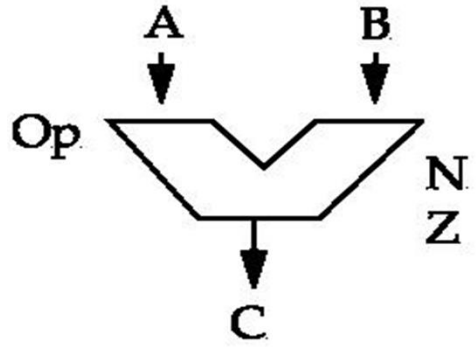
- Die size
  - yield, cost
- # of drawn FETS
  - design time, risk
- Levels on interconnect
  - cost, signal integrity
- Density (FETS/mm2)
  - yield, silicon efficiency
- Scalability
  - future costs, risk
- Verification time
  - NRE, functional first silicon
- % of chip active each cycle
  - silicon efficiency
- Power supply, noise margins
  - packaging cost, reliability
- Operating frequency
  - packaging costs, overhead
- W/Hz
  - battery life, power dissipation



VLSI Design Process

Simulation: The functional behavior of the design (or a parameter such as power) is determined by applying a set of excitation vectors to a circuit model.

VHDL (Hardware Description Language)



Specification

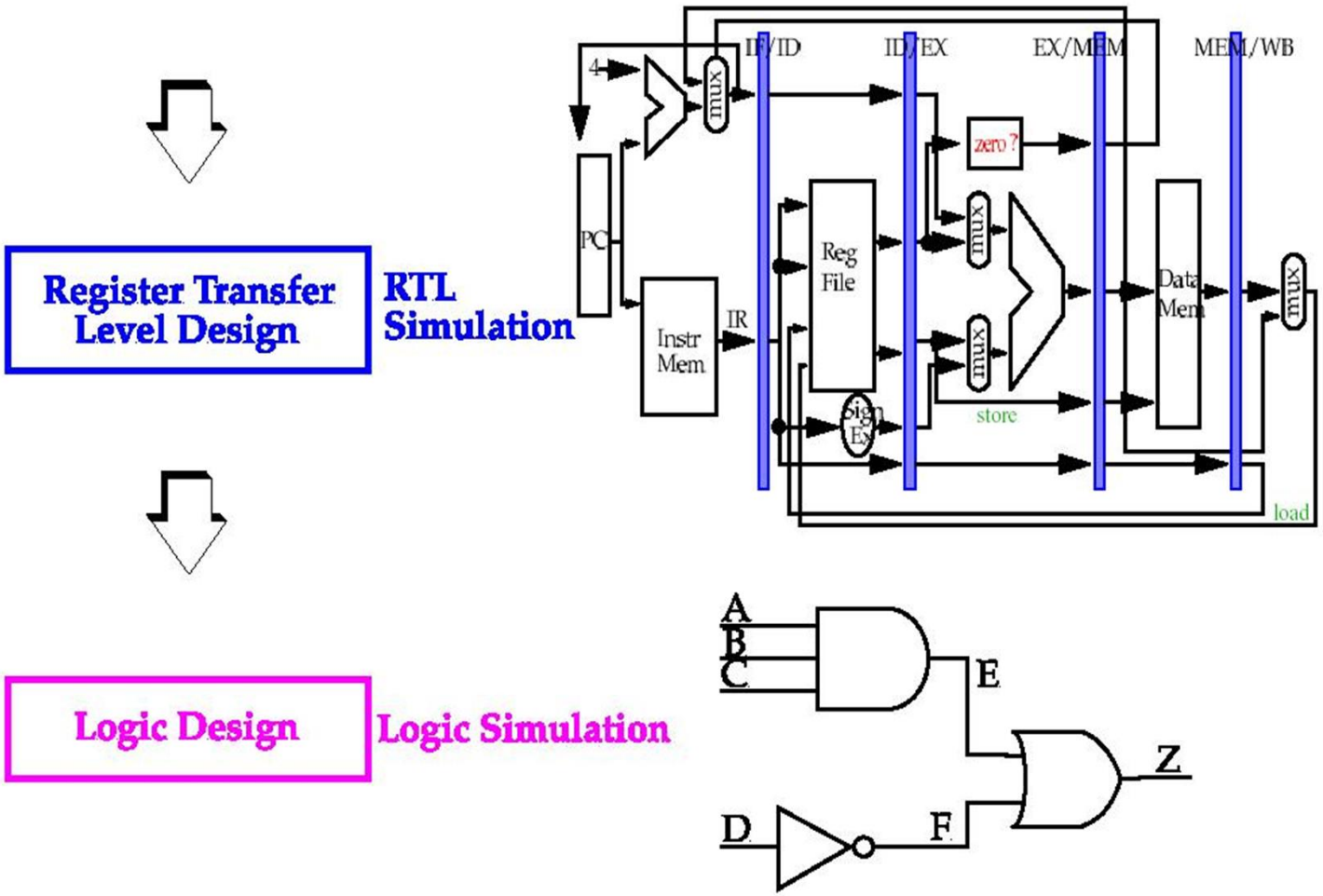


Functional Design **Behavioral Simulation**

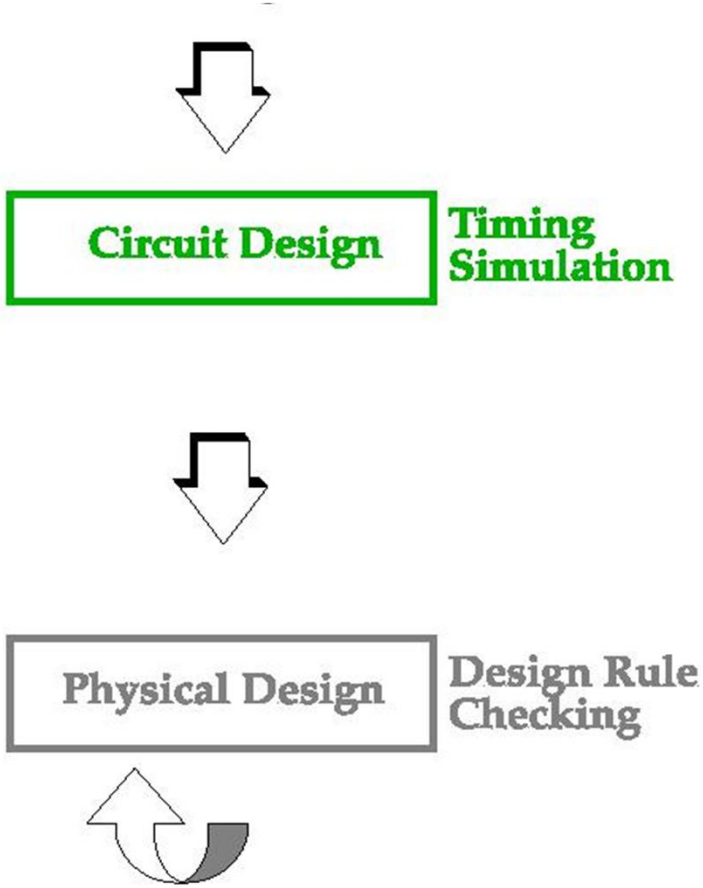
```
entity ALU32 is port(
    A, B: in bit_vector( 31 downto 0);
    C: out bit_vector( 31 downto 0);
    Op: in bit_vector( 5 downto 0);
    N, Z: out bit);
end half_adder;
```

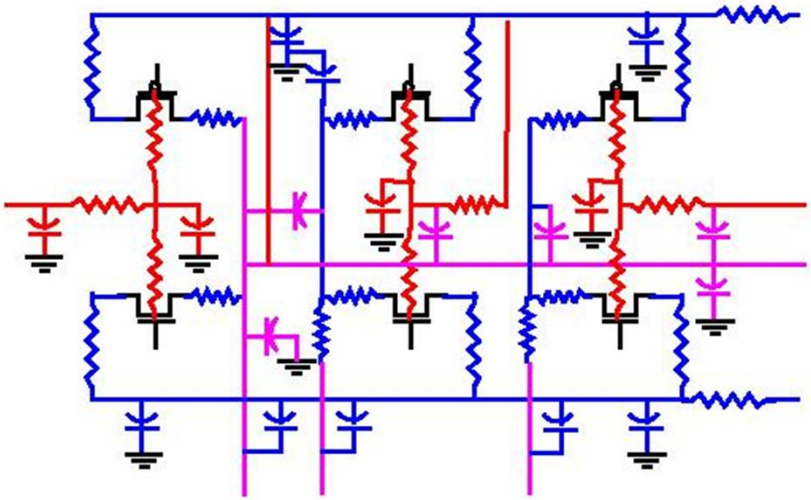


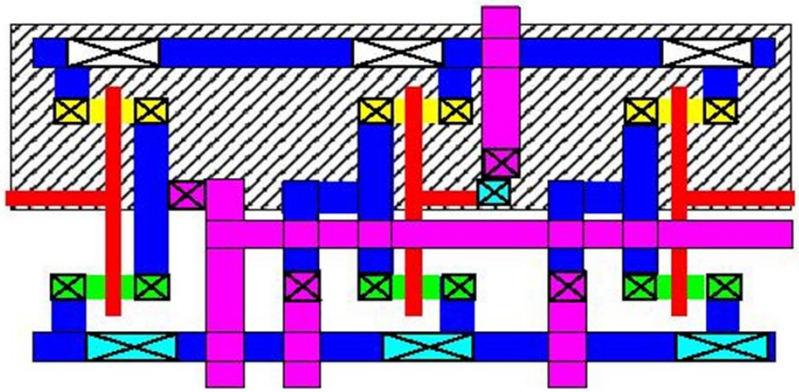

The VLSI Design Process



The VLSI Design Process





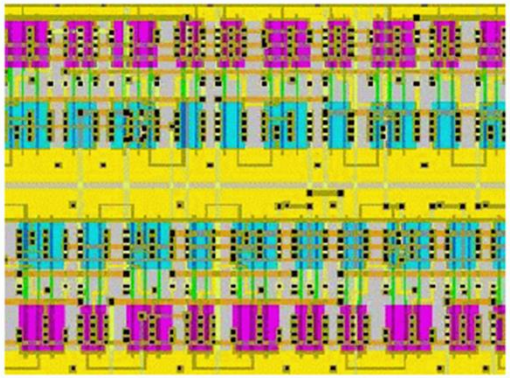




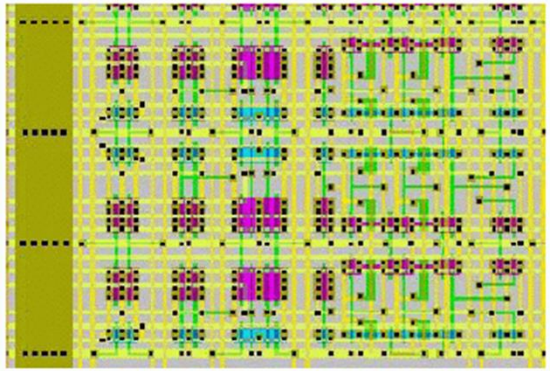
Implementation strategies

One could just start drawing transistors, but this quickly becomes hopeless with a chip of any size. Instead we'll use one of the following strategies:

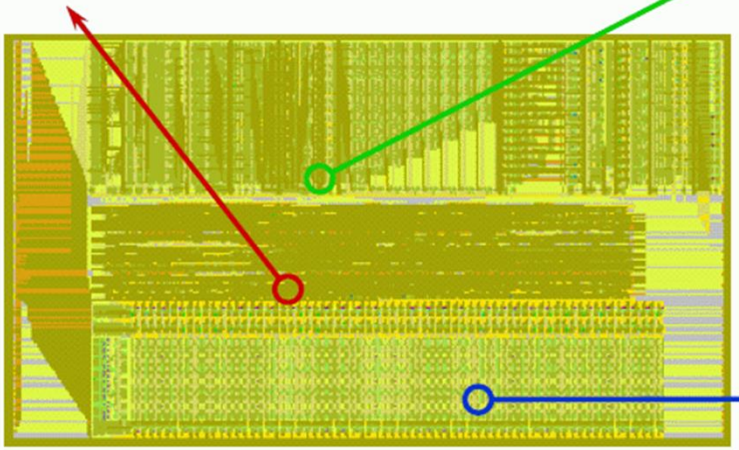
Standard cell: predefined gates, automatically placed and routed. In .5u → 10K fets/mm²



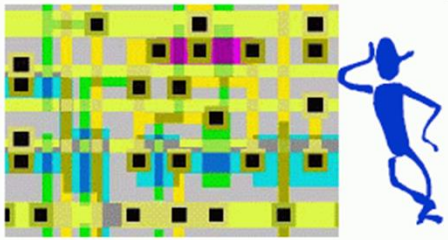
Full custom: custom "cells" meant to be stacked in columns to create N-bit wide datapath. Signals between columns routed across cells. In .5u → 25K/mm2







RAM Generator: one cell iterated many times perhaps surrounded by driver/sensing logic. Basic structure stays the same, only dimensions change. In .5u → 45K/mm² for multiport regfile



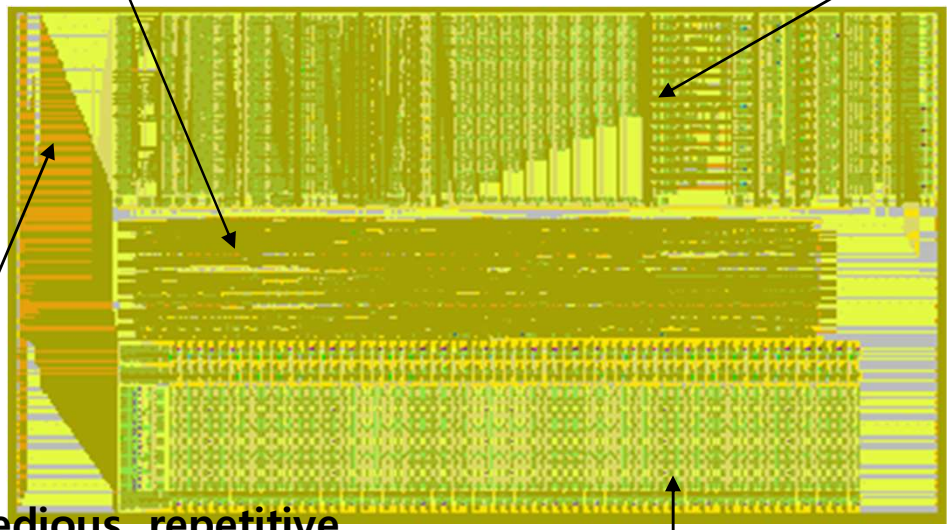


CAD Tools

◆organize ◆generate ◆verify

Standard-cell place and route for “random” logic.

Automated layout tools to ease the task of physical design; mask verification to ensure manufacturability.



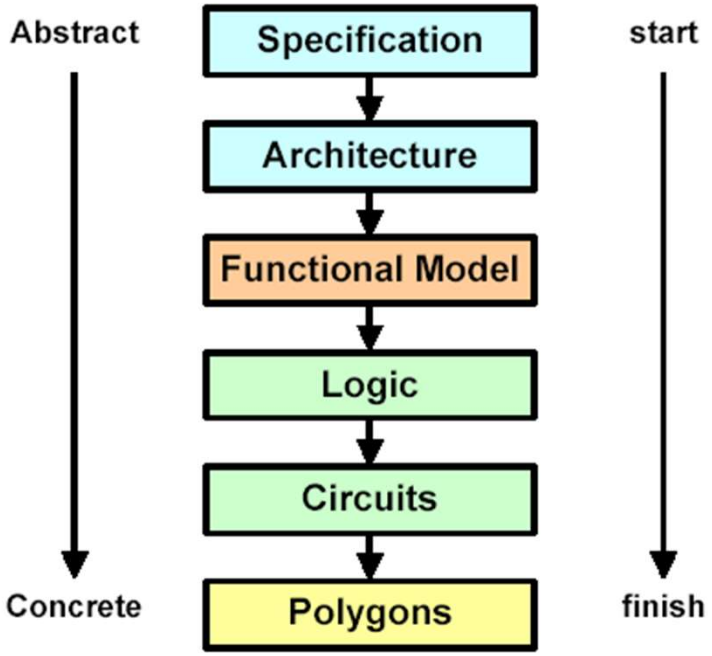
Tools to do the tedious, repetitive work such as routing,“tiling” a mosaic of building-block cells, or verifying that the layout and schematic match.

Circuit analysis programs predict circuit behavior at all the process corners. Gatelevel and behavioral simulators help you get it right the first time!



Levels of Design Abstraction

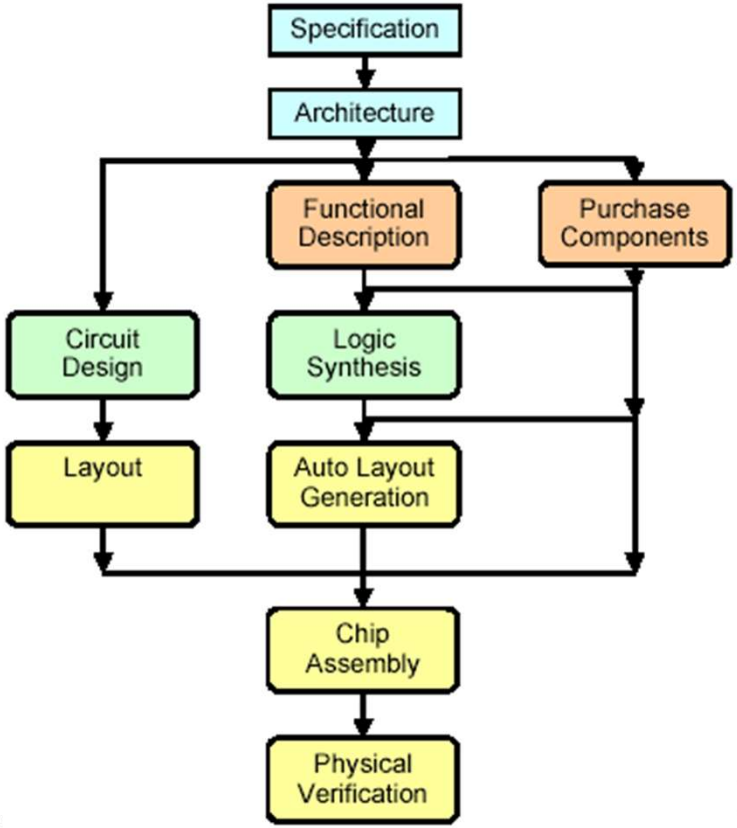
- Consider entire design at top levels only
- Increasing amounts of automation possible further down the stack





Design Methodologies and Flows

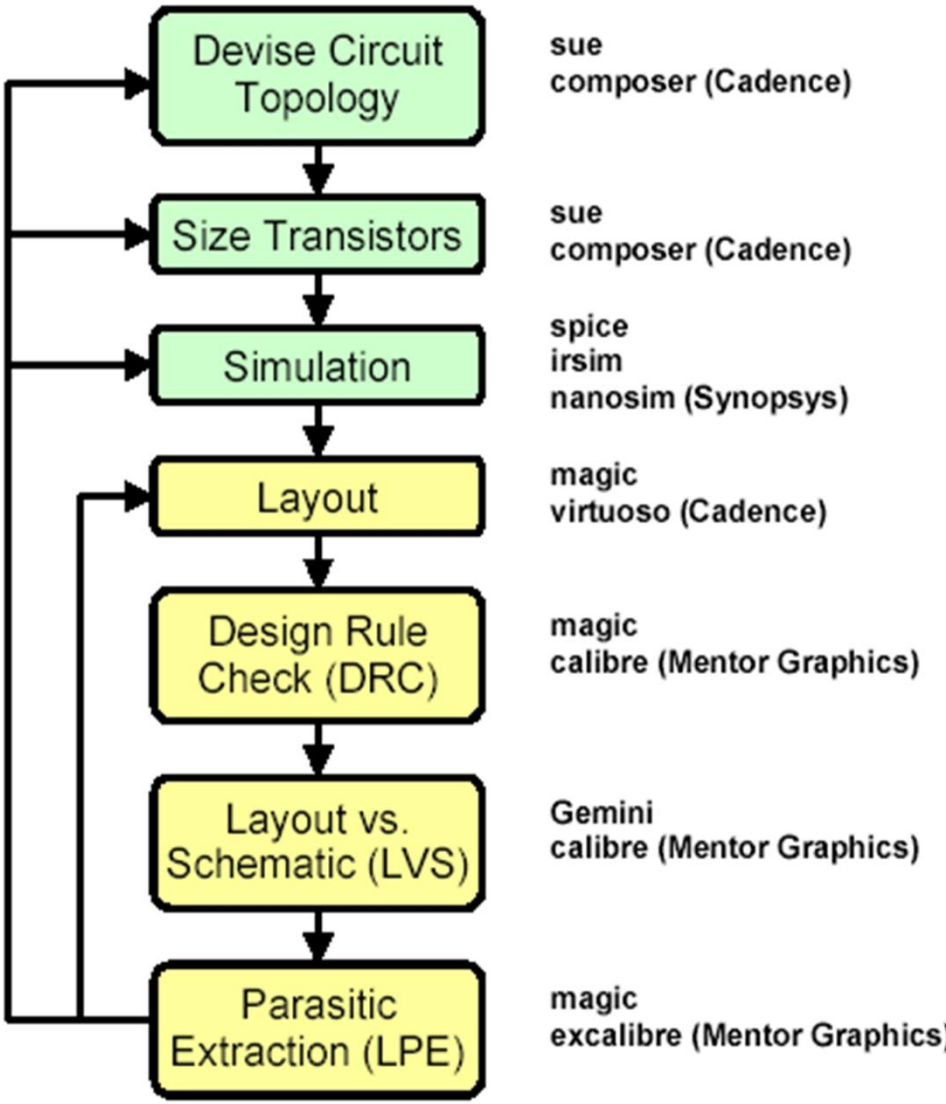
- Left fork: Full custom design flow
- Center fork: "ASIC" flow
- Right fork: System on Chip (SOC) flow





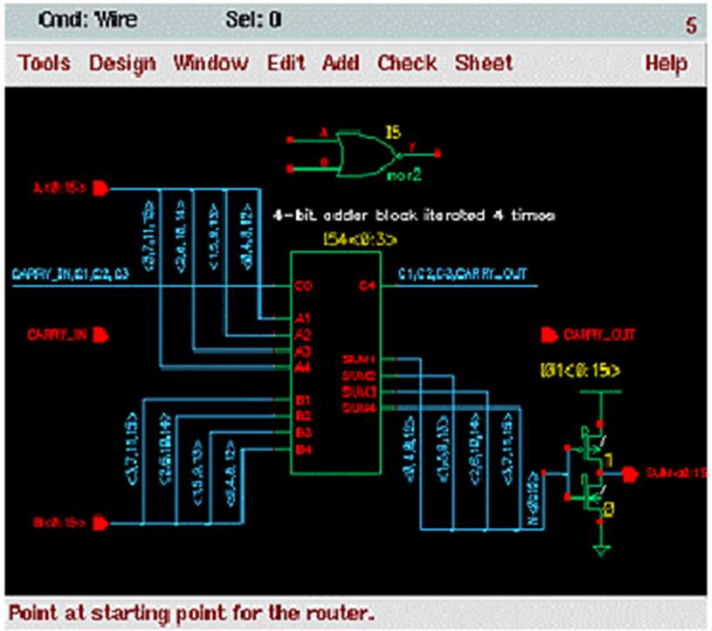
Full Custom Design Flow

- Has the best performance
- Is the most labor intensive



Schematic Capture/Simulation

- Circuit drawn at transistor, gate, and block level
- Blocks can be recursively placed inside one another
- Utility programs produce netlists for simulation tools

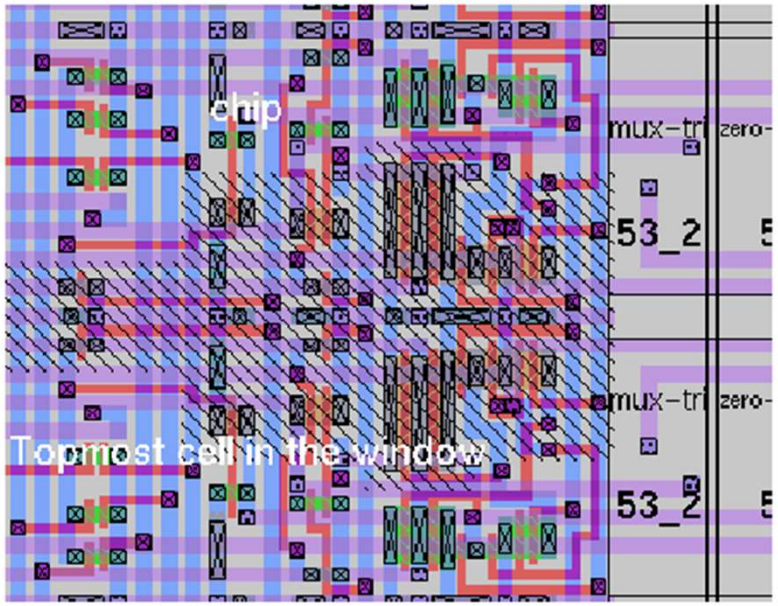


Cadence Virtuoso Schematic Composer



Layout

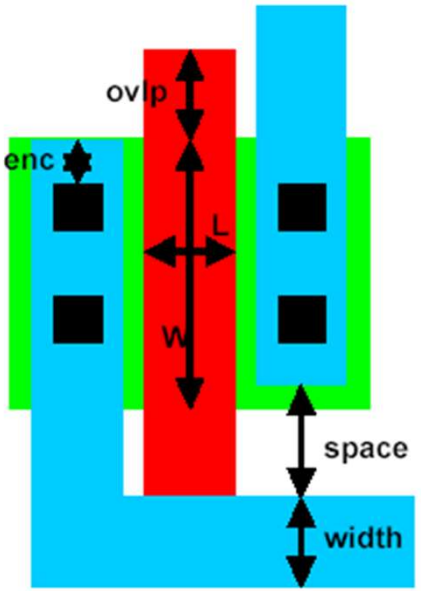
- Draw and place transistors for all devices in schematic
- Rearrange transistors to minimize interconnect length
- Connect all devices with routing layers
- Possible to place blocks within other blocks
  - Layout hierarchy should match schematic hierarchy



magic layout editor

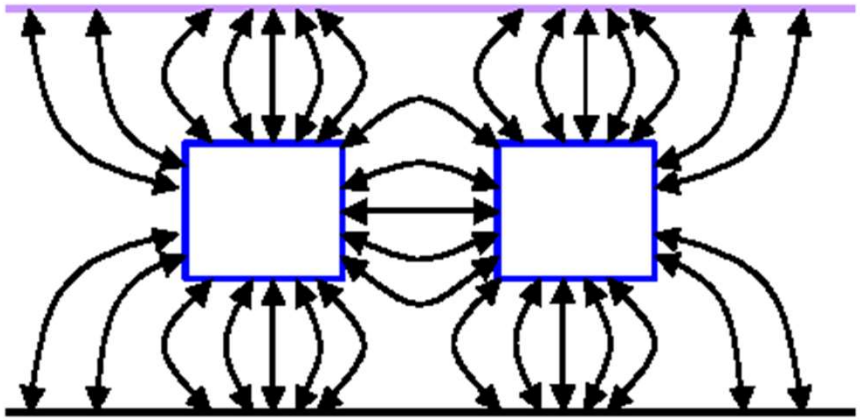
Design Rule Checking (DRC)

- Fab has rules for relationships between polygons in layout
  - Required for manufacturability
- DRC checker looks for errors
  - width
  - space
  - enclosure
  - overlap
- Violations flagged for later fixup





Layout Parasitic Extraction (LPE)

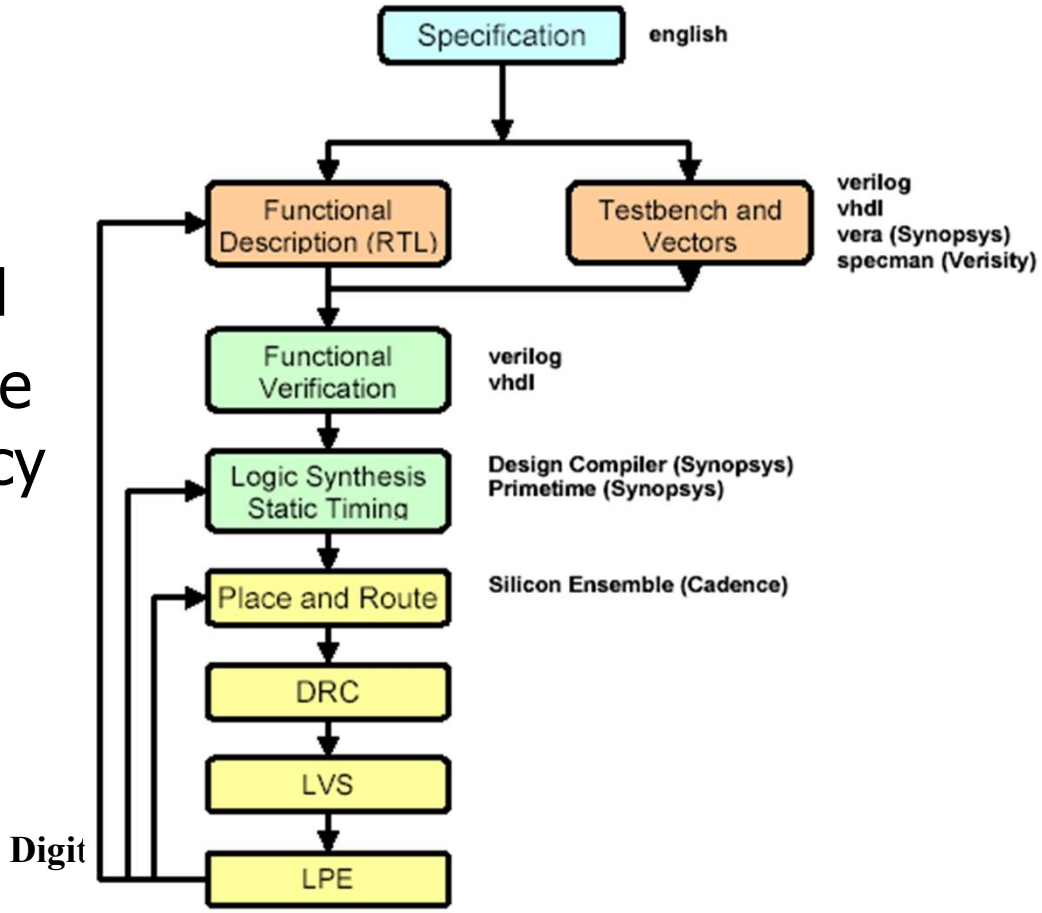


- Estimates capacitance between structures in the layout
- Calculates resistance of wires
- Output is either a simulation netlist or a file of interblock delays



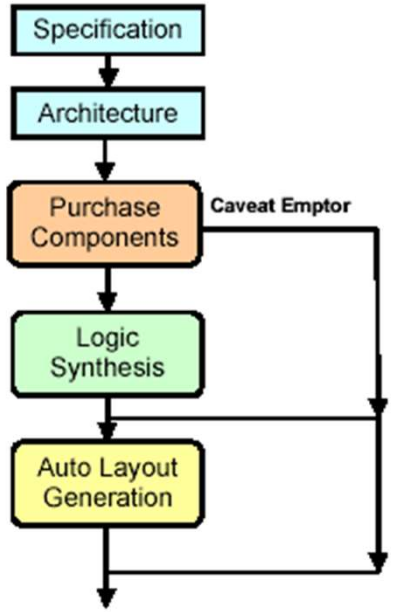
"ASIC" Design Flow

- Separate teams to design and verify
- Physical design is (semi-) automated
- Loops to get device operating frequency correct can be troubling



System on Chip Design Flow

- Can buy "Intellectual Property" (IP) from various vendors
- "Soft IP": RTL or gate level description
  - Synthesize and Place and Route for your process.
  - Examples: Ethernet, USB
- "Hard IP": Polygon level description
  - Just hook it up
  - Examples: XAUI Backplane driver, embedded DRAM
- Also: Standard cell libraries for ASIC flow





Chip Assembly

- Integrate blocks from previous steps into final layout
- Early floorplanning is key



Tapeout

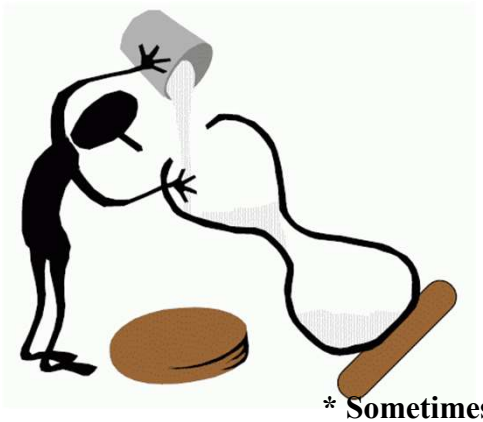
- Used to write 9-track computer tapes for mask making
- Final result of the design process for ICs before they are sent for manufacturing
- Now, Transfer polygons to fabrication company via ftp
- You're done! (Except for documentation, test, vector generation, device bringup, skew lots, reliability tests, burnin…)



VLSI Fact-of-Life #1: "So much to do, so little time"

You need a *design methodology* :

- **budget ($, speed, area, power, schedule, risk)**
- **low-level building blocks, high-level architecture**
- **specification***
- **behavioral design, verification***
- **logic design, verification**
- **layout, verification**





*** Sometimes skipped in the "old days"**



VLSI Fact-of-Life #2: "You can't reach in and fix it"

Notice that the word *"verification"* kept appearing in the previous slide.

Mistakes can be costly:

find bug(s)	?	?
reverify	1wk	?
new masks	3da	$350k
fab run	12wk	$1.5k/wafer
slip ship date		$$$

There's a lot that needs checking:

- circuit must operate at all "corners"
  - –verified at building block level
- logic must be correct, operate reliably
  - – verified at behavioral & RTL/gate level
- chip has to interoperate with system
  - –verified at behavioral level
- chip has to be manufacturable
  - –verified at mask level, at tester



VLSI Fact-of-Life #3: "You can't find all the bugs"

The key word here is "find":

- one can't explore the behavior of the circuit under all possible conditions
- some of the bugs arise from unanticipated interactions which, by definition, one never thinks of testing
- it's not clear when one is "done" looking for bugs! Time pressures mean that most searches stop too soon.

The trick is to choose some implementation rules that result in a circuit that is *correct by construction**.

For example:

- choose a simple clocking scheme
- module inputs must go only to fet gates
- disallow unclocked feedback
- make register t(clk-to-Q) > t(hold)+skew
- use poly only for local interconnect
- no diffusion wires
- etc., etc., etc.

* or at least avoids as many problems as possible!



VLSI Fact-of-Life #4: "Nobody's perfect"

Plan for what happens after you turn it on and nothing happens.

- provide lot's of *observability* and *controllability*. You'll need to localize and then find the bug.
- have a way to run the chip slowly and/or stop it without it burning up or loosing bits.



- figure out how to track down performance problems without relying on fast I/O (tester pins are slow!)
- leave room in the budget (time, $) for debugging.
- write and run your manufacturing tests before tape out.



Summary

• Digital integrated circuits have come a long way and still have quite some potential left for the coming decades

• Some interesting challenges ahead

– Getting a clear perspective on the challenges and potential solutions is the purpose of this programme

• Understanding the design metrics that govern digital design is crucial

– Cost, reliability, speed, power and energy dissipation



VLSI Design Flow

- The design starts with a given set of requirements
- Initial design is developed and tested against the requirements
- When requirements are not met, the design has to be improved
- If improvement is not possible or too costly, then a revision of requirements and its impact analysis must be done



VLSI Design Flow

- Y chart consists of three domains of representations: Behavioural, Structural and Geometrical layout
- The design flow starts from the algorithm that describes the behaviour of the target chip
- The corresponding architecture of the processor is first defined. It is mapped on to the chip surface by floorplanning
- The next design evolution in the behavioural domain defines FSM which are structurally implemented with functional modules such as registers and ALUs
- These modules are then geometrically placed onto the chip surface using CAD tools for automatic module placement followed by routing with a goal of minimizing interconnect areas and signal delays

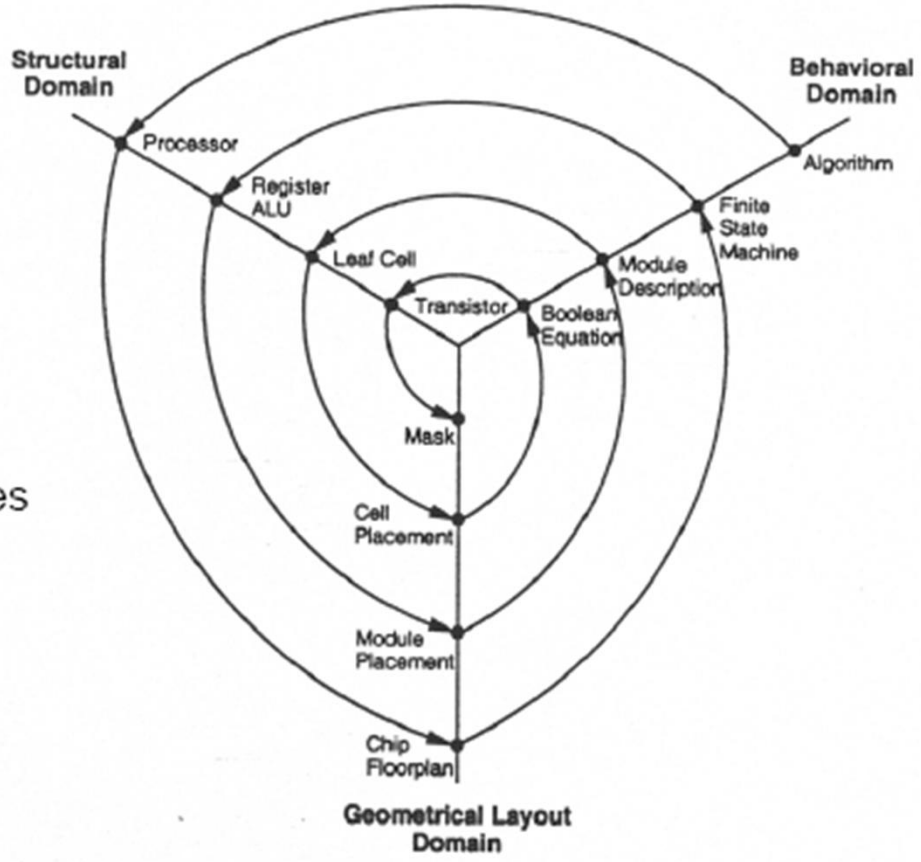


VLSI Design Flow

- The third evolution starts with a behavioural module description
- Individual modules are implemented with leaf cells. At this stage, the chip is described in terms of logic gates which can be placed and interconnected using a cell placement and routing program
- The last evolution starts with a detailed Boolean description of leaf cells followed by a transistor level implementation of leaf cells and mask generation
- In the standard cell design style, leaf cells are pre designed and stored in a library for logic implementation, effectively eliminating the need for the transistor level design



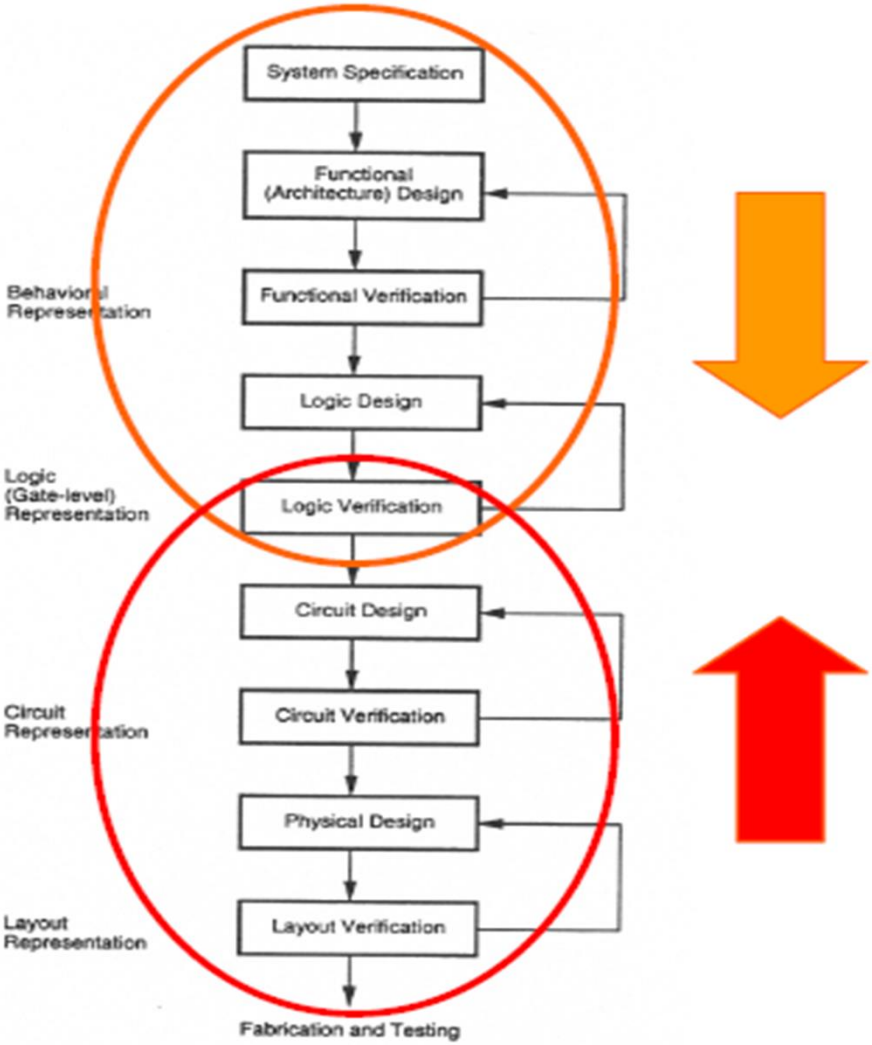
The Y-Chart



Notice: There is a need for structured design methodologies to handle the high level of complexity !



Simplified VLSI Design Flow



Top-down

Bottom-up



Structured Design Principles

Hierarchy: "Divide and conquer" technique involves dividing a module into sub-modules and then repeating this operation on the sub-modules until the complexity of the smaller parts becomes manageable.

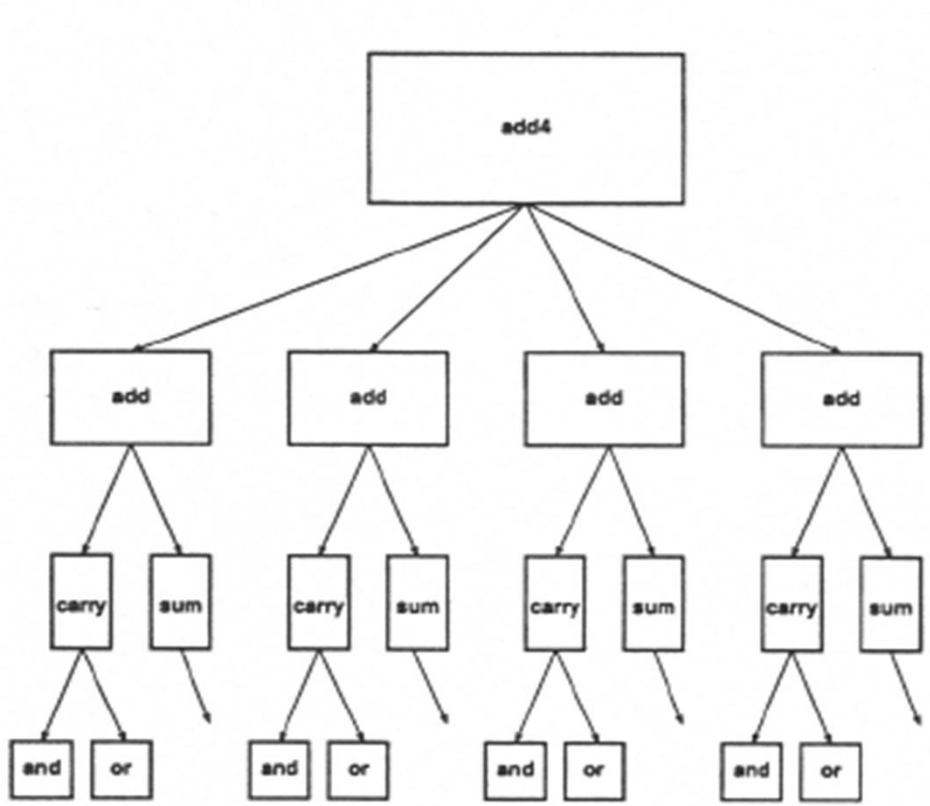
Regularity: The hierarchical decomposition of a large system should result in not only **simple**, but also **similar** blocks, as much as possible. Regularity usually reduces the number of different modules that need to be designed and verified, at all levels of abstraction.

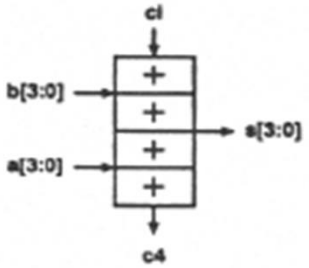
Modularity: The various functional blocks which make up the larger system must have **well-defined functions** and **interfaces**.

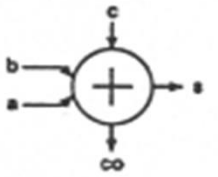
Locality: Internal details remain at the local level. The concept of locality also ensures that connections are mostly between neighboring modules, **avoiding long-distance connections** as much as possible.

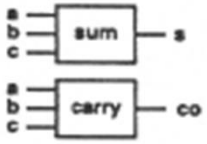


Hierarchy of a 4-bit Carry Ripple Adder





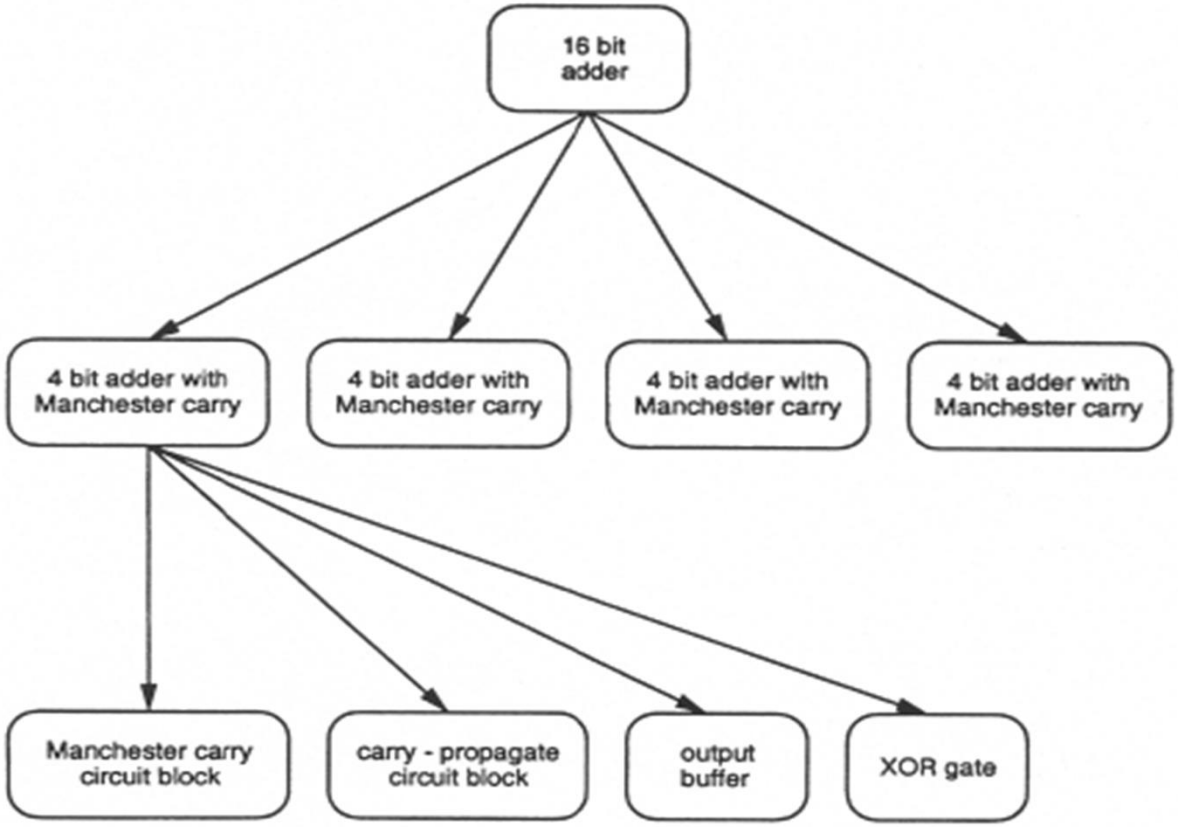




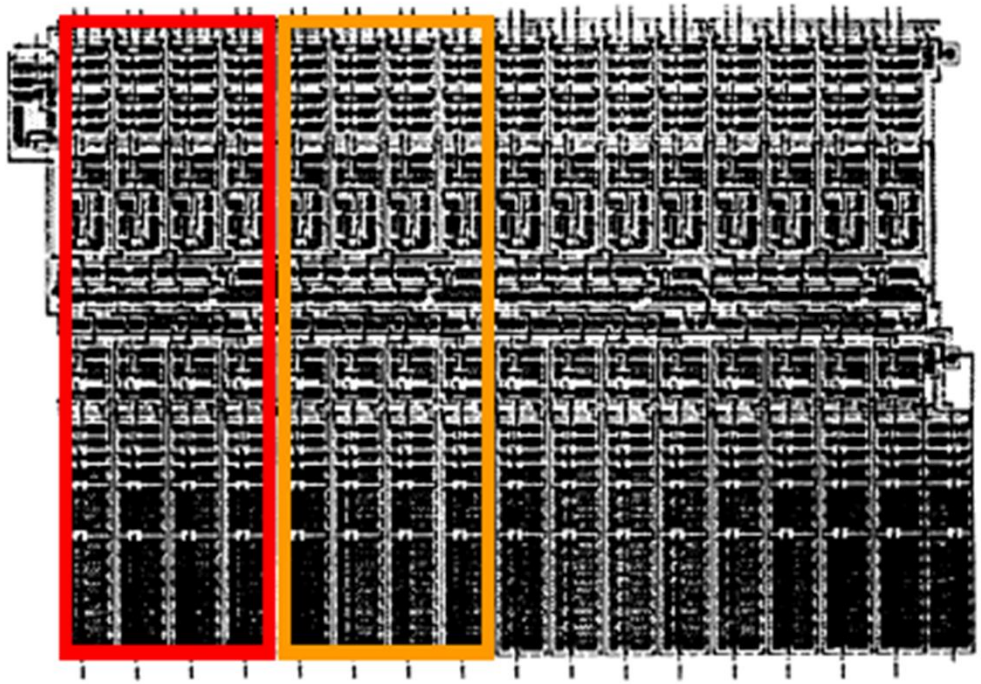




Hierarchy of a 16-bit Manchester Adder



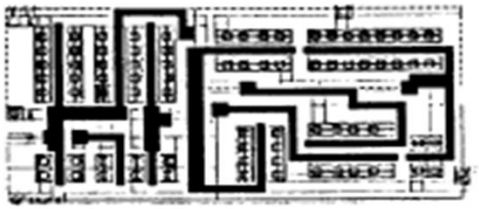
Hierarchy of a 16-bit Manchester Adder



16-bit adder
complete layout



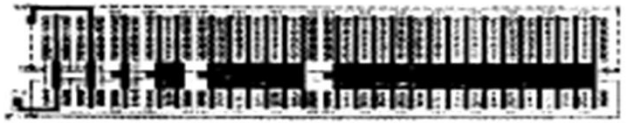
Hierarchy of a 16-bit Manchester Adder



Carry/propagate
circuit layout



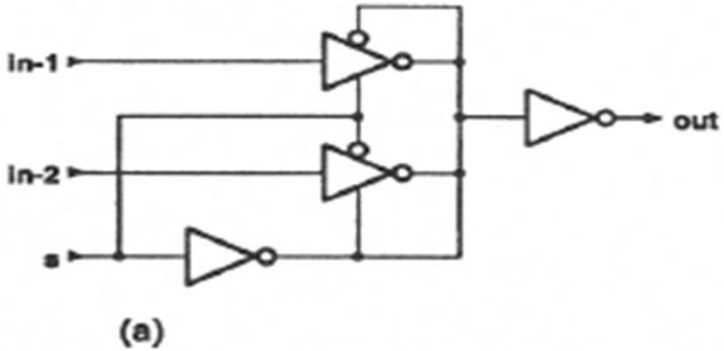
Manchester carry
circuit layout



Output buffer/latch
circuit layout



Regularity



2-input MUX

DFF

